Banksy: Evidence-Gated Repair of Garbled Text in AI-Generated Videos

Anonymous CVPR submission

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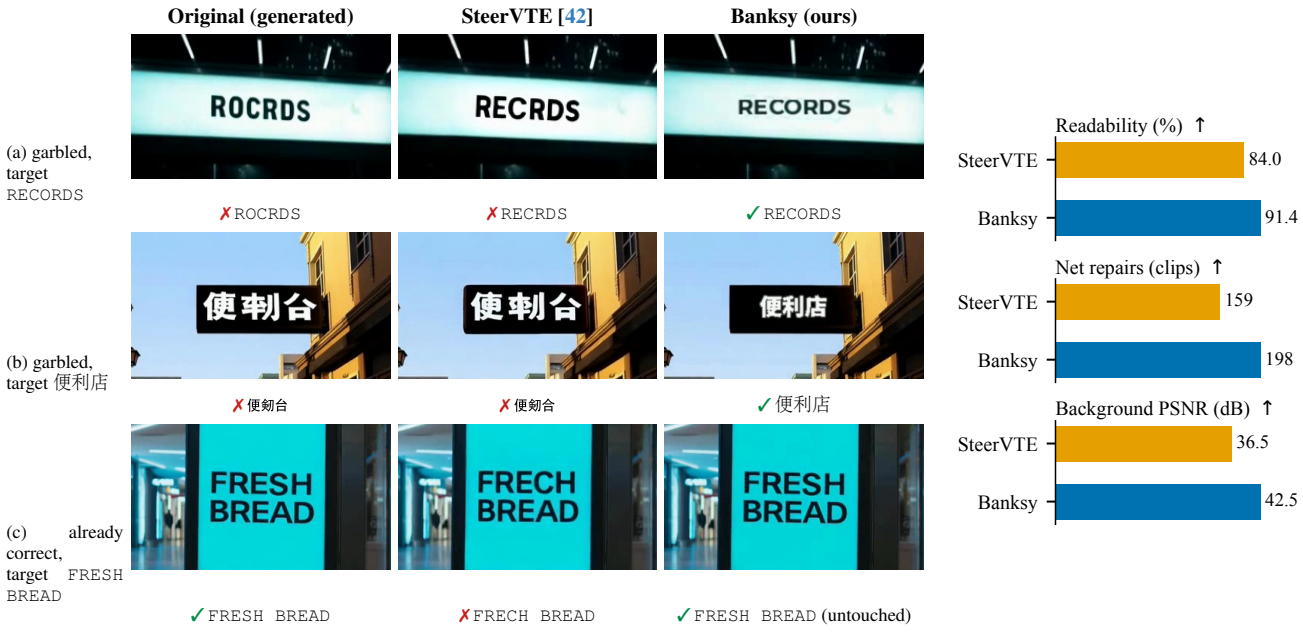


Figure 1. **Repairing garbled text in generated videos.** A generative video text editor rewrites the text region and often produces plausible but wrong characters (a, b) or breaks text that was already correct (c). Banksy edits only when evidence supports it, writes the target string deterministically, and leaves everything outside the text untouched. Below each crop: the transcription of frame 24 by the evaluation reader (✓ correct, ✗ wrong). Right: headline numbers on our 526-clip benchmark (Sec. 5.1). Examples (a, b) were chosen among clips where SteerVTE fails and Banksy succeeds; (c) is the only clip among the 46 that Banksy leaves untouched which SteerVTE breaks.

Abstract

Text-to-video generators still render garbled text on signs, screens and posters. We study repairing such text: given a generated clip, a target string and a hint box, make the text read correctly while leaving everything else as it was. Generative video text editors produce natural-looking glyphs, but they frequently keep or invent wrong characters, rewrite text that was already correct, and re-render the background. We present Banksy, a training-free, correctness-first repair pipeline built around three evidence gates. A preservation gate leaves a clip untouched when multi-frame OCR already reads the target; a localization gate lays the new text out on the old glyph instance when the evidence is strong and falls back to the tracked box otherwise; and an appearance gate enables a non-parametric source appearance transport (SAT) that carries glow and material from the old glyphs to the new ones only where it measurably helps. The target string itself is rendered deterministically, so its spelling cannot drift, and only pixels inside the text region change. On a benchmark of 526 generated clips spanning seven carrier materials in English and Chinese, Banksy reaches 91.4% readability against 84.0% for the state-of-the-art generative editor SteerVTE (198 vs. 159 net repairs; a pre-registered second reader from another model family confirms the advantage, 84.8 vs. 78.7%) and keeps the background intact (42.5 vs. 36.5 dB PSNR outside the text region). In a blind study, raters preferred SAT

over plain pasting for material fidelity in 87% of non-tied judgements. We also report, with pre-registered criteria where applicable, what did not work—a learned 1.3B refiner, glyph-only erasure, a carrier-fit check, reuse of correctly generated source glyphs, and baseline-aligned layout with glyph-matched fonts—and we characterise the appearance gap to generative editing that remains.

1. Introduction

Text-to-video models such as Wan [31], CogVideoX [41] and VACE [16] synthesise convincing street scenes, shop fronts and screens, yet the text inside them is often garbled: a record shop reads ROCRDS, a convenience store reads 便刻台 (Fig. 1). In our benchmark of generated clips, only 53.8% of the target strings are rendered correctly. For advertising, signage, education or any use in which the text carries meaning, a single wrong character is a hard failure, and re-generating the clip until the text happens to come out right is expensive and changes the shot. We therefore study *repair*: keep the generated clip, and make its text read correctly.

The natural tool is a generative video text editor [42], which regenerates the text region conditioned on rendered glyphs and on the style of the source. Such editors produce glyphs that blend into the scene, but on generated videos we observe three systematic problems. They write characters that look plausible but are wrong—FRESY BREAD, 小心坍滑, 洹影院—and on our benchmark the strongest of them, SteerVTE [42], still leaves 16% of the clips unreadable. They rewrite text that was already correct, turning FRESH BREAD into FRECH BREAD (Fig. 1c). And because the whole clip passes through the generator, the background changes as well: the PSNR outside the text region drops to 36.5 dB.

We argue that repair should be correctness-first and minimal, with a strict order of priorities: do not change what is already right; when a change is needed, change only the text; and recover the material of the old text only when there is evidence that this helps. Banksy implements this order with three evidence gates (Fig. 2). A preservation gate returns the clip unchanged when OCR on many frames already reads the target. A localization gate decides whether the new text can be laid out on the old glyph instance—its size, colour and baseline—or must fall back to filling the tracked box. An appearance gate decides whether to apply source appearance transport (SAT), a non-parametric operator that carries glow and material from the old glyphs to the new ones. Between the gates, the target string is rendered deterministically from a matched font and warped with carrier homographies, so its spelling cannot drift, and only pixels inside the text region are composited. The pipeline requires no training.

On 526 generated clips covering seven carrier materials in English and Chinese, Banksy raises readability from 53.8% to 91.4% (SteerVTE: 84.0%), with 198 net repairs against 159, and keeps the background intact (42.5 dB). We are equally explicit about what it does not do. Its glyphs are rendered, not generated: where both methods repair a garbled clip, SteerVTE's text is closer to the original appearance in 70% of the clips, and Banksy breaks 12 originally correct clips where SteerVTE breaks 6. We trace these gaps to their causes (Sec. 6.2) and report the alternatives we tried and rejected, including two modules aimed directly at the appearance gap that did not pass their pre-registered criteria (Sec. 5.7).

Our contributions are:

- **A repair task, benchmark and protocol.** 526 generated clips with hint boxes and targets, seven materials, two languages; metrics that separate repairs from breakages (net repair), background fidelity and collateral change, temporal stability, and blind human judgements. Development clips are disclosed and excluded in a separate report; the selective-editing study, the two appearance modules of Sec. 5.7 and the second-reader evaluation were pre-registered.
- **Banksy**, a training-free, evidence-gated repair pipeline: preservation and localization gates, deterministic glyph rendering anchored to the tracked carrier, and SAT behind an appearance gate.
- **An extensive analysis including negative results**: a learned 1.3B refiner trained on up to one million pairs does not improve appearance and costs readability and temporal stability; glyph-only erasure and a carrier-fit check fail their pre-registered criteria; reuse of the correctly generated source glyphs changes too few clips with our current segmentation, matching and gating to form a general module, baseline-aligned layout with glyph-matched fonts shows no overall benefit under our evaluation protocol, whose appearance metric is blind to glyph shape, so its perceptual effect remains unverified; and a glyph-locked material generator trained on rendered counterfactual pairs keeps the glyphs but does not transfer to generated video; and a failure taxonomy that locates the remaining appearance gap. Because the two systems fail on different clips, we also register and evaluate a verified selection between them: a reader independent of the scorer accepts SteerVTE's text where it reads correctly and composites it locally into the original, which reaches 96.4% readability with 0 broken clips (Sec. 5.8).

2. Related work

Scene text editing in images. Early editors decompose a text image into background, foreground style and content and re-synthesise it with GANs [17, 27, 28, 36]; STEFANN [28] generates missing characters in the font of an

observed one. Diffusion models brought text rendering and editing to arbitrary images through glyph conditioning [4, 15, 21, 32–34, 40, 43, 47]. These methods produce natural glyphs, but spelling remains a sampling outcome rather than a guarantee, and in video, frame-wise application produces flicker. A related line renders text first and harmonises it afterwards [37, 44]; our learned refiner (Sec. 5.6) belongs to this family.

Text in generated and edited videos. Text-to-video models [16, 31, 41] render legible text only occasionally: dedicated benchmarks find most open and commercial models unable to produce legible, temporally consistent text [11, 13, 45, 49], and image and video tokenizers already lose small text in reconstruction [35]. Generation-time remedies condition on glyphs [19] or fine-tune on synthetic text-rich clips [22]; they lower the rate of garbled text but do not repair a given clip. For editing, SteerVTE [42] injects line- and character-level glyph features and a VLM style embedding into a frozen VACE backbone and trains on a million synthetic and real triplets. VidScribe [46] evaluates text generation and localised editing in videos, including commercial models, and finds editing the weakest regime, with edits often corrupting text that should not change; ViTeX-Bench [5] evaluates video scene-text editing, including edit locality, and releases a 14B reference editor. All of these regenerate the text region. We instead repair generated text with a training-free pipeline that leaves already-correct text untouched, writes the target deterministically and changes only the text region, and we compare against SteerVTE as the strongest generative editor.

Text effects transfer. Statistics-based and example-based methods transfer the appearance of stylised text to new glyphs [25, 38, 39], using distance to the glyph boundary as the key coordinate. SAT follows this idea but fits a light-transport operator on the old glyphs of the same video and gates its use on measured reconstruction gain.

Video inpainting and consistent editing. Removing the old text is a video inpainting problem [3, 48]; we inpaint once on a reference frame with LaMa [30] and propagate the result along the tracked carrier, which avoids per-frame hallucination. Flow- and correspondence-guided attention improves the temporal consistency of diffusion-based video editing [6, 10]; we test a carrier-anchored attention bias of this kind on our learned refiner (Sec. 5.6).

Evaluating text. OCR systems [1, 7, 29] and, increasingly, vision-language models [2] are used to score generated text. We use a VLM reader for evaluation and a separate OCR inside the pipeline, report repairs and breakages separately, and complement automatic scores with blind human studies.

3. Task, benchmark and metrics

Task. The input is a generated clip V, a target string s and a hint box b that marks where the text is. In our benchmark the target is the quoted sign text of the generation prompt, and the hint box is produced automatically by OCR localisation and tracking of the generated text; the same box is given to every method, and SteerVTE uses it as its editing mask. The output is a clip $\hat{V}$ of the same size and length in which the text inside b reads s and nothing else has changed. Unlike scene text editing, the source text is usually not a valid word—it is a near miss produced by the generator—and in about half of the clips it is already correct, so "do nothing" is often the right answer. We call the task *repair* to stress both aspects.

Benchmark. We generated short clips of street scenes, shop fronts and screens with the text-to-video mode of VACE-14B [16], from 120 prompts that ask for a specific sign text on one of seven carrier materials, with ten seeds each (832×464 pixels, 49 frames at 15 fps). Of 1,178 usable clips, an OCR screen confirmed the requested text in only 24%. From the rest, an EasyOCR-based screen selected 526 clips as garbled; this screen is stricter than our evaluation reader, which judges 53.8% of them correct, so the benchmark contains both repair and do-nothing cases (exact selection rule to be documented). The 526 clips come from 117 prompts with up to ten seeds each, 40 target strings (20 English, 20 Chinese, from one to four words or characters, *e.g.* `BUS STOP`, `SALE 50%`, 小心地滑), and seven carrier materials: neon (93 clips), signboard (86), paper (96), glass (76), screen (65), metal (56) and wood (54); 295 clips are English and 231 Chinese. Every clip has a hint box per frame. In the evaluation job of Tab. 1, the reader (below) judges 283 originals (53.8%) correct and 243 garbled. Forty clips were looked at while developing the selective-editing components; they are listed in the supplement and excluded in a separate report.

3.1. Metrics

Readability. A vision-language reader, Qwen2.5-VL-7B [2], transcribes the text inside the hint box (15% padding) on 5 frames spread over the clip. A clip counts as *correct* if at least 4 of the 5 transcriptions match the target exactly after removing whitespace and line breaks, lower-casing, and folding traditional to simplified Chinese; readability is the share of correct clips, and NED similarity the mean normalised edit similarity over the 5 frames. The reader is independent of Banksy, whose gates use a different OCR system. All outputs that are compared are read in

the same job and software environment, and a second reader from a different model family re-reads the main comparison with the same protocol (supplement).[1]

Repairs and breakages. Readability alone hides two opposite effects, so we also count *fixed* clips (garbled → correct), *broken* clips (correct → wrong) and their difference, the *net* repairs.

Locality. *BG PSNR* is computed between output and input on all pixels outside the hint box dilated by half its height, over 49 frames. *Collateral change* is the mean absolute grey-level change, on the reference frame, inside the text neighbourhood (box grown by half its height) but outside both the old glyphs and the new glyphs (each dilated by 0.15 box heights); it isolates damage that is not part of writing the new text, and it requires the method's glyph mask, so we report it for Banksy only.

Appearance. The original glyphs are wrong but their material is the reference. On three frames, we estimate an ink mask inside the box for every video and compare, in CIELAB, the ink colour, the ink–ground contrast, the halo around the strokes (normalised by contrast), the edge softness and the shading inside the strokes with those of the original. Each feature is divided by its median over all method–clip pairs of the evaluation run and the five are averaged; lower is closer. The normalisation makes values comparable only within one run, which is why every table compares variants from a single run. The ink segmentation is not seeded, so re-scoring an identical video changes its distance by 0.47 (standard deviation per clip); we interpret only differences well above this noise (supplement).

Temporal stability. The carrier's frame-to-frame motion is estimated once per clip on the original (SIFT, RANSAC homography) and applied to every output, so camera motion cancels. *Flicker* is the mean absolute motion-compensated grey-level change inside the box, *jitter* the same for gradient magnitudes (stroke position), and *boundary jump* the change across the frame pairs at which a 17-frame chunked model switches chunks, relative to all other pairs (1 means no seam; used for the learned refiner in Sec. 5.6).

Human judgements. Blind pairwise studies show the original next to two versions in random left/right order, with the target known to the raters; questions are asked separately (*e.g.* material fidelity, artifacts, flicker), and ties are allowed. Study designs, sampling and rater quality control are in the supplement.

[1] We learned this the hard way: one version of the reader produced garbled transcriptions for about 8.5% of the inputs on one cluster and none on another; comparing outputs read in different environments inflated an early result (supplement).

4. Banksy

Banksy takes a generated clip $V = \{I_t\}_{t=1}^{T}$, a target string s and a hint box b (Sec. 3) and returns a clip $\hat{V}$ that differs from V only inside the text region. Figure 2 summarises the pipeline; all components are classical or pre-trained and nothing is trained for the task. We describe the steps in order and give thresholds and implementation details in the supplement.

4.1. Reading the text and tracking its carrier

We run an off-the-shelf OCR system (EasyOCR [14], CRAFT detection [1] and CRNN recognition [29]) on every frame and keep the boxes that overlap b. Boxes are grouped into text rows, and the frame whose rows best match s (normalised edit similarity) becomes the reference frame $t^\star$. The text is printed on, or emitted by, a physical *carrier*—a sign, a poster, a screen—so we track the carrier rather than the glyphs: SIFT keypoints [23] are matched between $I_{t^\star}$ and every I_t inside a window one text-box size larger on each side of the reference quad, and a RANSAC [9] homography H_t is estimated (ratio test 0.75, reprojection threshold 2.5 px, at least 10 inliers). When a frame fails, H_t falls back to the quad-to-quad homography between the OCR boxes. Everything that is placed on the carrier later—erased background, glyphs, masks—is placed once in the reference frame and carried to frame t by H_t, which removes per-frame jitter of the OCR boxes from the output.

4.2. Preservation gate

Generated text is often already correct, and any edit can only make such a clip worse. The preservation gate returns $\hat{V} = V$ when the OCR evidence across frames shows that the clip already reads s: the complete target is read in at least 85% of the frames, wrong readings occur in at most 5%, the reading lies inside the hint box, at most 10% of the frames contain extra boxes, and the reading of all boxes inside the hint box, concatenated in reading order, equals s in at least 90% of the correct frames. The last condition prevents skipping when the target appears next to other, garbled text (*e.g.*, a correct word above a garbled subtitle). The gate uses only the OCR of the pipeline, never the evaluation reader (Sec. 3.1), and when the evidence is insufficient the clip is edited: the gate trades recall for safety. In the final run it skips 46 of the 526 clips (9%), all of which the independent reader confirms as correct.

4.3. Localization gate and layout

If the old glyphs can be localised reliably, the new text should inherit their size, colour and baseline. We estimate a glyph *instance* in the reference frame: the ink mask (colour

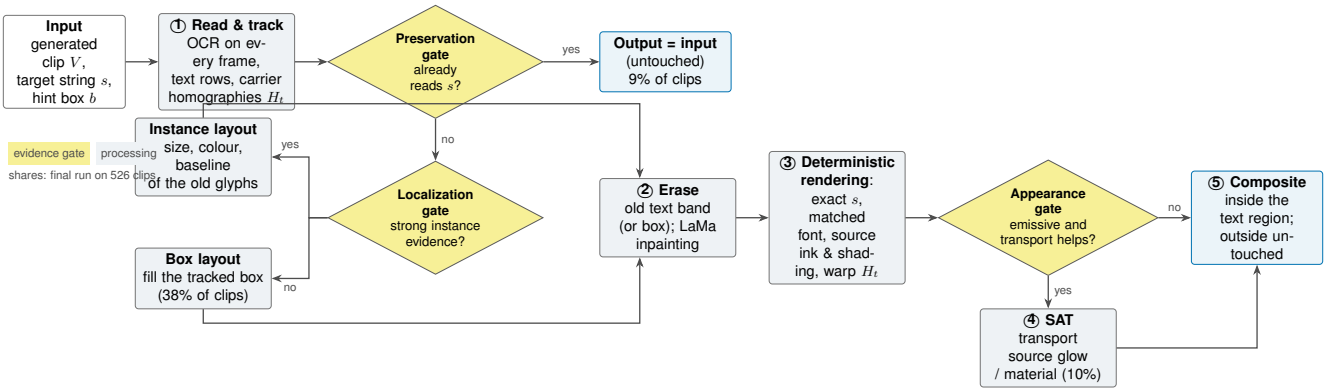


Figure 2. **Banksy overview.** Three evidence gates decide how much to change. The preservation gate returns the input unchanged when multi-frame OCR already reads the target; the localization gate chooses between laying the new text out on the old glyph instance and filling the tracked box; the appearance gate enables source appearance transport (SAT) only on emissive carriers where it measurably reduces the reconstruction error. The target string is always rendered deterministically, so its spelling cannot drift, and only pixels inside the text region are composited.

clustering inside the rows), the ink and ground colours, the letter height, the number of letters and the letter band. The localization gate accepts the instance only if it contains at least three letters (two for Chinese), the local contrast between ink and inter-letter ground exceeds 40 in CIELAB, the letter height is between 0.3 and 1.0 of the OCR box height, and the OCR rows did not require a fallback grouping. Otherwise the clip falls back to a *box layout* that fills the tracked box with legible text. In the final run, 168 of the 480 clips that Banksy edits (35%) take the box layout.

With an accepted instance, the target is laid out on the letter band: the cap height equals the measured letter height, the string keeps its natural aspect ratio in the selected font, and it is centred on the old text; a target wider than the old line may extend the line by up to $1.35\times$ before its height is reduced. The band is oriented like the OCR quad, which for most detectors is axis-aligned—a limitation we return to in Sec. 6.2. Stacked text keeps its rows: the target is split into lines following the word counts of the detected rows, and each line is laid out on its own row instance.

4.4. Erasure and deterministic rendering

Erasure. The old glyphs are removed once, in the reference frame: the ink mask (closed, and extended by the bright pixels of glowing text) is dilated by $\min(0.3h, \max(10, 0.18h))$ pixels for text height h, flat grounds are pre-filled with their median colour, and the hole is inpainted with LaMa [30]. The erased background is then carried to every frame by H_t, with a per-frame, per-channel gain fitted on a ring around the hole and smoothed over five frames, so that exposure changes follow without per-frame inpainting flicker.

Font matching. The new glyphs are rendered, not generated. The font is chosen from a library of 33 Latin and 8 Chinese families (variable fonts expanded over their weights) by comparing stroke statistics of the warped old ink mask with those of each candidate rendered in the same layout; script and display faces are admitted only when they are clearly better (distance below 0.6 of the best plain face). Because the target string is drawn by a font rasteriser, its spelling is exact by construction.

Photometry. The glyph colour follows the per-frame ink colour of the old text (running median over seven frames) modulated by a low-frequency shading field estimated from the old strokes in canonical text coordinates, and glowing text receives a soft halo. The canvas is warped to every frame with H_t and composited over the erased background.

4.5. Appearance gate and source appearance transport

Rendering gives the right letters, but on emissive carriers—neon, lightboxes, screens—the old glyphs carry glow, bloom and inner structure that a flat fill cannot reproduce. SAT transports this appearance from the old glyphs to the new ones without regenerating the region. Let B be the erased background, I the reference frame and $d(x)$ the signed distance of pixel x to the boundary of the old glyphs, in units of the letter height. On the old text we fit, per distance bin k and per channel, a multiplicative-plus-additive light-transport operator

$$I(x) \approx m_k\, B(x) + a_k, \qquad k = k(d(x)), \tag{1}$$

over the support $d \le 0.6$, and apply it to the new glyphs with their own distance field, $\hat{I}(x) = m_{k(\hat{d}(x))} B(x) + a_{k(\hat{d}(x))}$. On the old text, this family reconstructs the source

better than an additive-only operator, a distance-indexed replacement or a flat fill (cross-validated CIELAB error 17.2 vs. 25.9, 27.6 and 29.7; best on 45 of 52 clips; supplement), which is why we keep both terms.

SAT is applied only behind an *appearance gate* with two conditions: the text must be emissive (ink brighter than its ground, or a detected glow on the carrier), and the operator must generalise across the text—fitted on one half and tested on the other, in both directions, its error must stay below 0.75 of the flat-fill error. In the final run, SAT is enabled on 50 clips (10%); it is never applied to box-layout clips, whose instance evidence is weak by definition.

4.6. Compositing

The output differs from the input only where the erasure mask or the new glyphs (with halo) are non-zero; elsewhere the input pixels are copied, so the only residual outside the text region is the re-encoding of the video.

5. Experiments

Protocol. Table 1, Tab. 8 and every number in Secs. 5.1, 5.2 and 5.7 come from one front-end pass over the 526 benchmark clips with the final system and the variants of Sec. 5.7, which share the OCR, tracking and instance estimation and are therefore paired clip by clip; these outputs, the original clips and SteerVTE's outputs were read by the same reader in one evaluation job (Sec. 3.1). Every other table comes from its own front-end pass and its own evaluation run, named in its caption (Tabs. 2, 6 and 10 were evaluated on a second machine whose decoded crops are byte-identical to the first; supplement); because the appearance distance is normalised within each run, it is compared only within a table. Paired differences are reported with 95% bootstrap confidence intervals over clips [8] (10,000 resamples) and readability differences with exact McNemar tests [24]. Forty clips were inspected while developing the selective-editing components (Sec. 5.5); Tab. 1 reports all clips and, separately, the 486 clips that were never inspected.

Baseline. SteerVTE [42] is the strongest published video text editor; it adds glyph and style conditioning to a frozen VACE-14B backbone [16] and is trained on one million triplets. We ran the released model on every clip with the hint box as editing mask and the target as text, at the same resolution and length as the input. Its outputs are generated end to end, so they are evaluated as they are.

5.1. Main results

Table 1 compares Banksy with SteerVTE and with the unedited clips. Banksy makes 91.4% of the clips readable, 7.4 points more than SteerVTE (70 clips are correct only with Banksy, 31 only with SteerVTE; McNemar $p = 1.3 \times 10^{-4}$), and turns 210 of the 243 garbled clips into correct ones. It is not free of damage: it breaks 12 of the 283 originally correct clips where SteerVTE breaks 6, so its advantage in net repairs (198 vs. 159) is smaller than in repairs alone. The picture is unchanged on the 486 clips outside the development set (91.2% vs. 84.4%; 177 vs. 144 net repairs).

The two methods trade different qualities. Banksy leaves the background intact—its PSNR outside the text region is 6.0 dB higher (95% CI [5.8, 6.3]; higher on 97.7% of the clips)—and its strokes jitter less (edge jitter 7.86 vs. 8.30), because every glyph is placed once on the tracked carrier. SteerVTE's text region flickers less (1.66 vs. 1.79) and, above all, its text looks more like the original: its appearance distance is 1.83 against 2.57 for Banksy. We analyse this gap in Sec. 6.2.

Figure 3 breaks readability down by carrier material. Banksy is ahead of SteerVTE on all seven materials; the largest gains are on signboards (+15.1 points) and neon (+8.6), where SteerVTE most often keeps the garbled characters, and the smallest on metal (+1.8) and wood (+1.9). By language, Banksy reaches 91.5% on the 295 English and 91.3% on the 231 Chinese clips (SteerVTE: 84.1% and 84.0%). Figure 4 shows randomly sampled examples, including two failures of Banksy (公园 on neon, 花店 on glass); Fig. 5 shows selected clips that only Banksy repairs, a case about five times as frequent as the reverse (41 vs. 8 clips outside the development set under both readers).

A second reader. Readability depends on the reader, so we registered a re-reading of the outputs of Tab. 1 by readers from other model families, with identical crops, frames, prompt, decoding and scoring, and a decision rule fixed in advance: the advantage counts as reader-robust only if the main second reader, InternVL2.5-8B (InternViT vision encoder, InternLM2.5 language model), finds Banksy more readable than SteerVTE with $p < 0.05$ (Tab. 2). It does: 84.8% against 78.7% (74 vs. 42 discordant clips, McNemar $p = 3.8 \times 10^{-3}$; 185 vs. 153 net repairs), although the margin is smaller than under the first reader and, split by language, significant only for Chinese (83.5% vs. 75.3%, $p = 0.02$; English 85.8% vs. 81.4%, $p = 0.11$). InternVL is stricter with every method and agrees with the first reader on about 91% of the clip–method verdicts. A third reader, Pixtral-12B, cannot read Chinese—it invents unrelated text on every Chinese crop—and was registered for the 295 English clips only; there it finds no advantage (77.6% vs. 80.3%, $p = 0.37$). Most of its errors on Banksy's outputs are correct transcriptions followed by invented text (9.4% of the frames, against 6.8% for SteerVTE), which exact matching counts as wrong; when a frame only has to contain the target (an exploratory criterion defined after the reads), Pixtral finds the two methods level (88.8% vs. 87.5%). The

Table 1. **Main results** (526 clips; one front-end pass and one evaluation job for every row). Readability: share of clips whose text the reader transcribes exactly in at least 4 of 5 frames. Fixed / broken: garbled → correct and correct → wrong transitions relative to the original as read in the same job; net = fixed − broken. Appearance: glyph-agnostic material distance to the original text (normalised within this run; lower is closer). BG PSNR: outside the dilated text region. Flicker and jitter: motion-compensated, inside the text box (grey levels). The lower block excludes the 40 clips inspected while developing the selective-editing components.

	Readability ↑	Fixed ↑	Broken ↓	Net ↑	NED sim. ↑	Appearance ↓	BG PSNR ↑	Flicker ↓	Jitter ↓
All 526 clips									
Original (generated)	53.8	–	–	–	0.820	–	–	1.98	9.18
SteerVTE [42]	84.0	165	**6**	159	0.946	**1.83**	36.49	**1.66**	8.30
Banksy	**91.4**	**210**	12	**198**	**0.959**	2.57	**42.52**	1.79	**7.86**
486 clips outside the development set									
Original (generated)	54.7	–	–	–	0.822	–	–	1.97	9.12
SteerVTE [42]	84.4	150	**6**	144	0.945	**1.85**	36.40	**1.65**	8.24
Banksy	**91.2**	**189**	12	**177**	**0.958**	2.60	**42.50**	1.79	**7.86**

Table 2. **Readability under different readers** (the outputs of Tab. 1; identical crops, frames, prompt, decoding and scoring). W/L: clips correct only with Banksy/ only with SteerVTE, with the exact McNemar p. Agree: share of clip–method verdicts equal to the first reader's (Tab. 1). The re-read uses the first reader on the second readers' hardware.

Reader	n	Original	SteerVTE	Banksy	W/L (p)	Agree
Qwen2.5-VL-7B (A40; Tab. 1)	526	53.8	84.0	91.4	70/31 (1.3×10^{-4})	–
Qwen2.5-VL-7B, re-read (V100)	526	53.8	84.0	91.8	70/29 (4.6×10^{-5})	99.9
InternVL2.5-8B (V100)	526	49.6	78.7	84.8	74/42 (3.8×10^{-3})	91.1
Pixtral-12B, English only (V100)	295	60.0	80.3	77.6	27/35 (0.37)	88.2
Qwen2.5-VL-7B (A40), English only	295	62.0	84.1	91.5	40/18 (5.4×10^{-3})	–

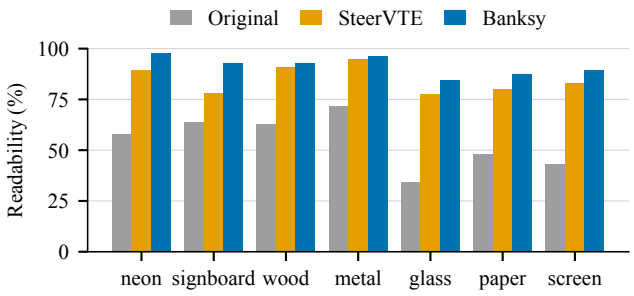


Figure 3. **Readability by carrier material** (all 526 clips; 54 to 96 clips per material).

second readers ran on different GPUs from the first reader, with byte-identical crops; the first reader re-reading the outputs on those GPUs reproduces 99.6–100% of its verdicts per method, so the hardware does not explain these differences.

5.2. Edit locality

Figure 6 visualises where each method changes the video. Because SteerVTE passes the whole clip through its generator, its difference to the original covers the entire frame, including edges of buildings and furniture far from the text; for Banksy, the difference is confined to the text region, and the residual elsewhere (1–2 grey levels) is the re-encoding of the output video. Inside the neighbourhood of the text, we additionally measure the collateral change—the change outside both the old and the new glyphs—which is 1.68 grey levels for Banksy on the 480 clips it edits; it cannot be computed for SteerVTE, whose glyph coverage is unknown.

5.3. The evidence gates

Preservation gate. The gate skips 46 clips, and the reader confirms all 46 as correct (no false skip; the pre-registered limit was 2%). Its effect on breakage is measured in the selective-editing pass (Tab. 6), where it is the only difference between B0 and B1: it restores 2 of the 16 originally correct clips that B0 breaks and loses none (16 vs. 14 broken clips, readability 91.3% vs. 91.6%). Its recall is low by design: it fires on 46 of the 283 clips the reader judges correct, because the pipeline's OCR rarely reads stylised originals—neon italics, extruded letters, condensed faces—with the required consistency. All 12 originally correct clips that Banksy breaks are such cases (Sec. 6.1).

Carrier anchoring and style-matched rendering. Table 3 ablates the rendering components on an earlier front end. Pasting the target into the per-frame OCR box makes the text readable but nearly doubles flicker relative to the original clip (3.12 vs. 1.98), because OCR boxes jitter from frame to frame. Anchoring everything on the tracked carrier removes most of it: carrier-space erasure contributes most (-0.93), tracking -0.32 and anchored masks with ink smoothing -0.05, and the three together reduce flicker by 1.45, more than the sum of their parts, to the level of SteerVTE (1.67 vs. 1.66) while lowering jitter well below it (7.12 vs. 8.30). Readability does not suffer (89.0 vs. 87.1, $p = 0.064$). Matching the font to the old strokes adds 2.6 points of readability (D1; $p = 0.044$) once candidates are

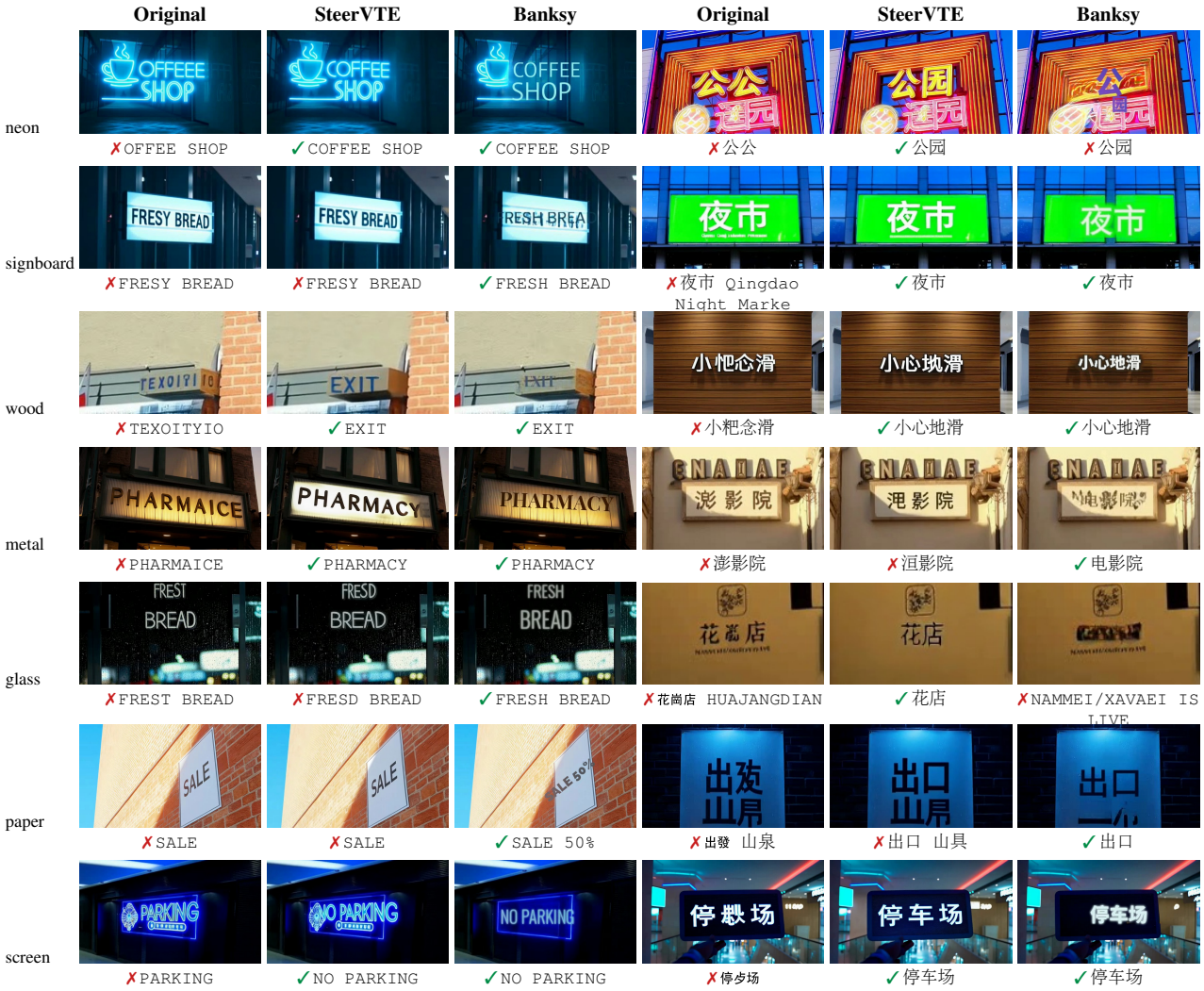


Figure 4. **Qualitative comparison on randomly sampled clips** (one English and one Chinese clip per material, drawn with a fixed seed among garbled originals outside the development set; not curated). Each cell is the same crop of frame 24; below it, the reader's transcription (✓ correct, ✗ wrong). SteerVTE produces more natural glyphs but frequently keeps or invents wrong characters; Banksy writes the target but its glyphs often look pasted (see Sec. 6.2).

rendered in the layout of the original—an earlier version compared single-line candidates with two-line originals and systematically picked the thinnest faces. Per-frame ink and a shading field (D2) bring the appearance closest among the rendered variants; imitating edge blur and grain (D3) makes text both less readable and less similar, and was dropped. T4 + D2 constitutes the *box policy*, the fallback of Banksy.

The localization gate. Laying the text out on the old glyph instance should look more faithful than filling a box, and on held-out clips it does: it leaves half as much of the old ink visible (residual 0.21 vs. 0.42 of the original). On all 526 clips, however, the first instance-aligned front end was 5.5 points less readable than the box policy (85.2 vs. 90.7; Tab. 4). Size and colour were not the cause—forcing the box policy's colour pipeline or a minimum letter height left the gap unchanged (supplement)—but its confidence-based tracking state machine was: replacing it by carrier tracking recovers most of the gap on the clips where an instance is found (90.8 vs. 91.3), and the remaining clips are those where the instance estimate itself is unreliable. The localization gate routes exactly these clips to the box policy, which yields a front end that is as readable as the box policy (91.1 vs. 90.7; 27 vs. 25 discordant clips) while keeping the instance path's lower residual (0.23) and lowest jitter (7.66). Two re-readings of the same outputs differ by about one point (90.3 vs. 91.3 in a second job), so we claim parity with the box policy, not superiority. A subset of 100 clips

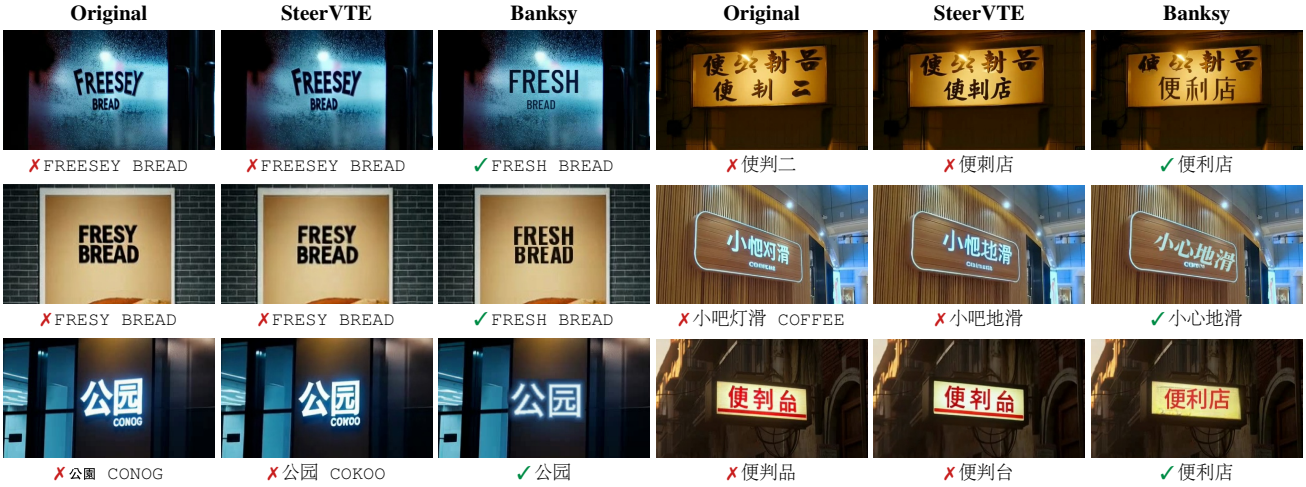


Figure 5. **Clips that only Banksy repairs** (selected by hand). The examples are chosen among the 41 clips outside the development set that both readers judge wrong in the original and in SteerVTE's output and right in Banksy's; the reverse case—right with SteerVTE and wrong with Banksy under both readers—occurs in 8 clips (failure modes in Fig. 9). Each cell is the same crop of frame 24, with the first reader's transcription below (✓ correct, ✗ wrong).

Table 3. **Rendering components** (earlier front end, 526 clips). Top: carrier-anchored temporal components; bottom: style-matched rendering. Appearance is normalised within each block. The box policy used by Banksy's fallback is T4 + D2.

	Read. ↑	Flicker ↓	Jitter ↓	Appear. ↓
Original	53.8	1.98	9.18	–
SteerVTE	83.8	1.66	8.30	–
Naive paste (per-frame OCR box)	87.1	3.12	11.66	–
+ carrier tracking (T1)	87.3	2.80	10.28	–
+ carrier-space erasure (T2)	88.6	2.19	9.27	–
+ anchored masks, ink smoothing (T3)	87.1	3.06	11.47	–
+ all three (T4)	89.0	1.67	7.12	–
SteerVTE	83.8	–	–	1.39
Naive paste	86.9	–	–	1.89
+ font matching (D1)	89.5	–	–	1.83
+ per-frame ink, shading field (D2)	89.4	–	–	1.82
+ edge blur and grain (D3)	87.6	–	–	2.01

Table 4. **Front-end variants and the localization gate** (526 clips, one evaluation job). W/L: clips correct only with the variant / only with the box policy (McNemar p). Residual: share of old ink still visible, relative to the original, on 24 held-out clips.

	Read. ↑	vs. box (W/L, p)	Jitter ↓	Appear. ↓	Resid. ↓
Original	53.8	–	9.18	–	1.00
SteerVTE	84.0	33/68, <.001	8.30	1.50	–
Naive paste	86.7	16/37, .005	12.53	2.38	–
Box policy (T4+D2)	90.7	–	7.80	2.27	0.42
Instance layout, state-machine tracking	85.2	32/61, .003	7.68	2.50	0.21
Instance layout, carrier tracking	86.9	34/54, .042	7.68	2.44	–
+ localization gate (Banksy front end)	91.1	27/25, .89	7.66	2.33	0.23

had suggested a gain of one point for an earlier gated variant (95 vs. 94) that did not replicate on the full set (88.4 vs. 91.3)—a reminder that subset gates are optimistic.

5.4. Source appearance transport

Where the gate enables SAT. In the pass analysed here, the appearance gate enables SAT on 53 of the 526 clips (in the run of Tab. 1, on 50): 23 of 93 neon clips, 11 of 65 screens, 9 of 86 signboards, 8 of 56 metal and 2 of 54 wooden signs, and none on glass or paper. Most rejections are clips without an accepted instance (203), followed by non-emissive text (133), no cross-validated gain (104) and gain in only one direction (33). Every enabled clip has a cross-validated error ratio below 0.75 in both directions, and the 15 rejected clips closest to the threshold lie between 0.756 and 0.795.

Table 5. **SAT on the clips where the appearance gate enables it** (53 of 526; paired, same front-end pass). Δ: SAT − paste only, with 95% bootstrap CI.

	Paste only	+ SAT	Δ [95% CI]
Readability (%)	98.1	100	+1 clip
Appearance ↓	2.97	1.95	-1.02 $[-1.70, -0.48]$
Flicker ↓	2.25	2.40	$+0.15$, $p < 0.001$
Jitter ↓	8.62	9.42	$+0.81$, $p = 0.001$
BG PSNR (dB)	43.01	43.02	n.s.

Effect. On the enabled clips (Tab. 5), SAT reduces the appearance distance by a third (-1.02; better on 42 of

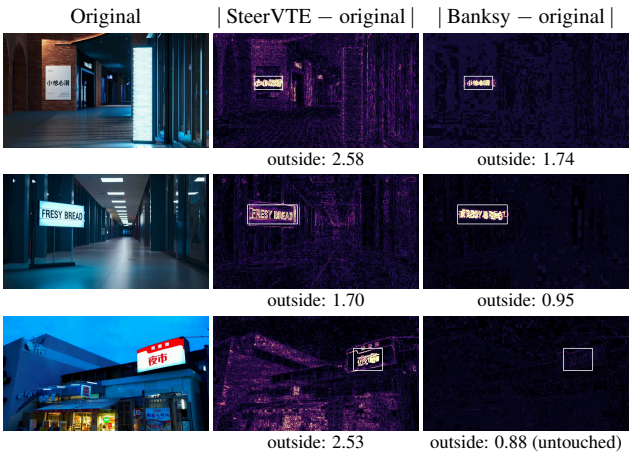


Figure 6. **Edit locality.** Absolute grey-level difference to the original frame (clipped at 24; white box: text region). SteerVTE re-renders the whole frame, so residual edges appear everywhere; Banksy changes only the text region. The numbers give the mean absolute difference outside the dilated text region. Banksy's residual there (1–2 grey levels, block artefacts of the output codec) is the video re-encoding floor, visible alone in the bottom row, where the preservation gate leaves an already-correct clip untouched while SteerVTE still alters it.

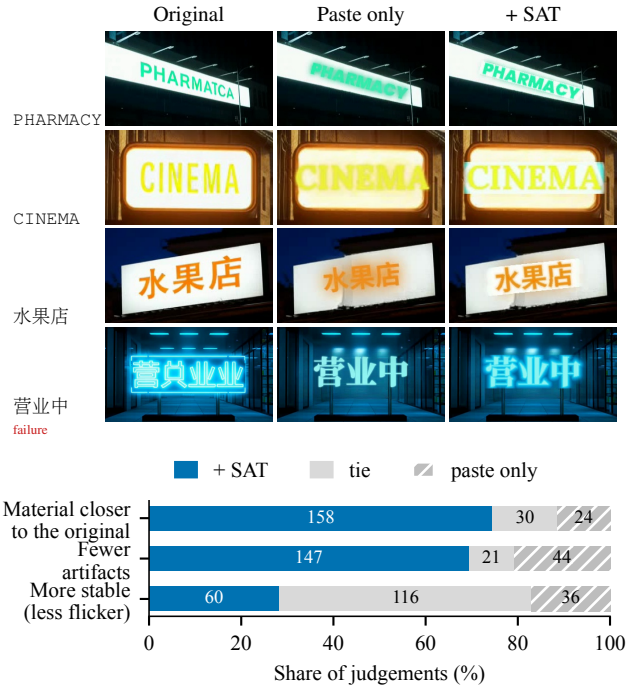


Figure 7. **Source appearance transport.** Top: SAT carries glow and material of the old glyphs to the new ones (rows 1–3: clips that raters judged better with SAT on both material and artifacts; row 4: a neon clip that all raters judged to have more artifacts with SAT). Bottom: blind pairwise study on all 53 SAT-enabled clips (4 raters, 212 judgements per question; Sec. 5.4).

53 clips) for every material—screens 3.02→1.33, neon 3.03→2.02, wood 4.12→2.21—without hurting readability or the background. It costs temporal stability: flicker rises by 0.15 and jitter by 0.81, because the transported appearance follows a per-frame distance field. On the full benchmark the readability is unchanged (90.5 vs. 90.3%).

Human judgement. Because the appearance distance is only a proxy, we ran a blind study on all 53 enabled clips (Fig. 7): raters saw the original between the paste-only and the SAT version in random order and answered three separate questions. Of five raters, one gave the same side on all three questions for 98% of the clips (including the flicker question, which asks which version is worse) and was excluded by a rule (identical answers on at least 80% of the clips) that we fixed only after seeing the data; we therefore also report the material question with all five raters. With four raters (212 judgements per question), SAT was judged closer to the original material in 158 cases against 24 (30 ties; win rate 0.87, 95% CI [0.77, 0.94] by bootstrap over raters and clips; 48 vs. 3 clips by majority), and as having fewer artifacts in 147 against 44 (0.77 [0.66, 0.87]); with five raters the material win rate is 0.84 [0.74, 0.92]. Agreement is moderate (Gwet's AC1 [12] 0.51 for material, 0.61 for artifacts), and the left/right position had no effect after balancing (53% left, $p = 0.27$). The measured flicker increase was not perceived: 55% of the flicker answers are ties, the remaining answers follow the material answer (a halo effect), and they correlate negatively with the measured flicker change ($\rho = -0.34$). The appearance distance agrees with the material votes only weakly ($\rho = 0.28$, $p = 0.049$) and not with the artifact votes, so it cannot replace human judgement.

Failures and their cause. Raters judged six clips to have clearly more artifacts with SAT, four of them neon (Fig. 7, last row): a bright rectangular plate behind the new letters. A first hypothesis—that the new glyphs reach distances never observed on the old ones—was rejected (the share of unobserved distances is at most 2.3%), and tapering the operator to identity at the edge of its support had no effect. The plates coincide instead with the canvas on which the new glyphs' distance field is computed: the canvas leaves only 0.04–0.21 letter heights of margin, so the halo, which should extend to 0.6, is cut at the canvas border. With an untruncated distance field the straight edges disappear in our diagnosis, and computing the distance field on a padded canvas reduces flicker by 0.08 and jitter by 0.37 (both better on 47 of 53 clips, $p < 0.001$) with no significant change in appearance—about half of SAT's flicker cost and 40% of its jitter cost. Because these 53 clips were used to diagnose the failure, the padded version is not part of the

evaluated Banksy and requires validation on unseen clips.

5.5. Selective, structure-preserving editing

Inspecting the outputs of the first full evaluation, we found that the most frequent problem of Banksy was not that its glyphs lacked material but that it changed things it should not have: already-correct text, sign frames, and the structure around the letters. We therefore tested three components that make editing more selective, each added to B0 (Banksy without the preservation gate) on its own, with acceptance criteria fixed before the run (supplement): the preservation gate of Sec. 4.2 (B1), *glyph-only erasure* (B2), and a *carrier-fit check* (B3).

Glyph-only erasure (B2). Instead of inpainting the whole letter band, B2 erases only the old glyphs and their own glow, estimated with a mixture model of carrier, ink and halo colours, and protects sign frames, long bars and the boxes of other text. It is active on the 325 clips with an accepted instance—in 117 of them it falls back to band erasure minus the protected structures, because the surroundings themselves look like strokes (*e.g.* neon stripes)—and its erased area is a median 0.64 of the band area. Preregistered criterion: collateral change decreases ($p < 0.05$) and readability drops by at most one point.

Carrier-fit check (B3). B3 estimates the extent of the carrier from colour, lightness and texture boundaries and declares a clip *layout-infeasible* if the target, at the character density of the source, would have to shrink below 0.7 of the source height to fit; such clips are left unedited and counted as failures. Criterion: the infeasible flags are confirmed by manual inspection.

Results. Table 6 reports all variants from one front-end pass, in which the 99 two-row clips were re-run after a compositing fix that affects all variants alike, and from one evaluation run for all rows. The preservation gate meets its criteria (no false skip; two fewer broken clips; readability not lower) and is adopted, which makes B1 the Banksy of Tab. 1. Glyph-only erasure reduces collateral change on all clips (-0.090, 95% CI $[-0.16, -0.02]$, $p = 0.016$) but not significantly once the development clips are removed ($p = 0.24$), and it costs 7.6 points of readability: the instance ink mask is not a complete glyph mask, so parts of the old strokes survive next to the new ones, and OCR boxes of the garbled text itself were sometimes protected as "other text". A second version that completes the old glyphs by colour still left remnants in three of nine diagnosed clips, which suggests that this component needs a learned stroke segmentation. Of the 24 clips flagged infeasible by the carrier-fit check, manual inspection confirmed 5, was undecided on 3 and rejected 16 (OCR missed letters of the source, or two-row signs were treated as one row), so B3 is rejected as well; a stricter version that only flags clips whose source stably shows a whole-word part of the target behaved correctly on the diagnosed clips but has not been run at scale. All three together (B4) inherit B2's readability loss. The B1 of this pass and the Banksy of Tab. 1 are the same configuration in two independent front-end passes, so their numbers differ slightly (the pipeline's GPU OCR is not bit-reproducible, and the appearance distance is normalised per run).

5.6. Does a learned refiner help?

Rendered glyphs look pasted, so the obvious remedy is to let a generative model harmonise them with the scene while keeping their shapes. We invested heavily in this route before settling on the training-free pipeline, and report it in full because its outcome shapes our design.

Refiner. The refiner is a video diffusion transformer initialised from Wan2.1-T2V-1.3B [31] that takes the pasted clip as extra input channels (zero-initialised) and is trained as a flow-matching bridge [18, 20] from the paste to the target; inference starts from the paste, in the spirit of SDEdit [26], and takes 30 Euler steps on 17-frame chunks of 384×384 crops. A glyph loss compares Sobel edges of the decoded prediction with the target inside a band around the glyph (weighted by the signed distance to the glyph boundary). Training pairs come from three sources: a Blender data factory with six carrier surfaces and passing occluders, which renders every sample twice (lit composite and flat paste); 562 real pairs obtained by erasing correct text from generated clips and re-pasting it; and up to 1.15 million image pairs built the same way from AnyWord-3M (63% LAION, 31% Wukong, 6% street scenes). All refiners were trained on H100-class GPUs (GH200/H200) for 20k steps.

More data is not the lever. On a 296-clip garbled subset, refiners trained with 0, 10k, 100k and 1M image pairs reach 78.4, 75.3, 76.4 and 76.4% readability, all below the paste they start from (80.7%), and their appearance does not improve with data (supplement). A diagnostic on 1,500 held-out pairs explains why: on the very task they were trained for—turning the paste back into the original—all of them move away from the original, they do not even memorise pairs seen twelve times, and they fail to learn a systematic difference (originals have thicker strokes than our pastes in 60% of the pairs). The supervision was inconsistent: the paste uses our font, the target uses the generator's, and the glyph loss, which dominates the objective, effectively teaches the model to erase our strokes. Making the supervision glyph-consistent—pasting the original's own ink mask instead of our font—is the first recipe that moves toward

Table 6. **Selective, structure-preserving editing** (pre-registered; 526 clips; one front-end pass, in which the 99 two-row clips were re-run after a compositing fix, and one evaluation run for every row: all rows read together, every metric computed over all rows at once). B0 is Banksy without the preservation gate; each component is added to B0 separately and then all three together (B4). Collateral: change outside the old and new glyphs in the text neighbourhood. The verdict applies the pre-registered criteria; development clips are excluded in the right block. The B1 of this pass and the Banksy of Tab. 1 are the same configuration in two independent front-end passes and evaluation runs.

	All 526 clips									486 clips (no development clips)		
	Read. ↑	Fixed	Broken	Net	Appear. ↓	BG PSNR	Collateral ↓	Flicker	Verdict	Read. ↑	Net	Collateral Δ vs. B0
B0: Banksy w/o preservation gate	91.3	214	16	198	2.52	42.97	1.55	1.76	–	91.2	178	–
B1: + preservation gate (= Banksy)	**91.6**	214	**14**	**200**	2.39	42.97	1.52	1.79	adopted	**91.4**	**179**	−0.031 [−0.06, −0.01]
B2: + glyph-only erasure	83.7	181	23	158	2.43	43.06	1.46	1.79	rejected	84.4	145	−0.036 [−0.09, +0.03]
B3: + carrier-fit check	88.8	200	15	185	2.39	42.95	1.54	1.77	rejected	88.9	167	−0.004 [−0.01, −0.00]
B4: all three	82.9	173	19	154	**2.21**	43.05	**1.43**	1.82	rejected	83.3	140	−0.062 [−0.12, +0.01]

Table 7. **Does a learned refiner help?** The 1.3B bridge-flow refiner applied on top of Banksy's front end (paste only) and of the box policy, with and without the carrier-anchored attention bias ($\lambda = 2$). 526 clips, one evaluation job; W/L against the same front end without the refiner. Boundary jump: change at the refiner's 17-frame chunk seams relative to other frame pairs (1 = no seam).

	Read. ↑	W/L	Flicker ↓	Jitter ↓	B. jump ↓	Appear. ↓
SteerVTE	84.2	–	1.66	8.34	0.93	1.58
Banksy front end (paste only)	91.3	–	1.77	7.69	0.84	2.49
+ refiner	87.8	3/21	1.96	8.22	2.26	2.45
+ refiner + attention bias	88.6	3/17	2.00	8.41	2.06	2.33
Box policy (paste only)	91.4	–	1.77	7.83	0.84	2.35
+ refiner	87.6	2/22	1.99	8.46	2.31	2.31
+ refiner + attention bias	88.4	3/19	2.03	8.68	2.10	2.24

the original on held-out pairs and it cuts the image-domain readability loss from 26 to 4 points, with appearance better than the paste. On videos, however, it still does not exceed the paste (best variant: 79.7 vs. 80.7%).

System-level effect. Table 7 applies the best refiner to the full benchmark. On top of Banksy's front end it costs 3.5 points of readability (21 clips broken against 3 repaired; $p < 0.001$), increases flicker and jitter, and introduces seams at its chunk boundaries (boundary jump 2.26 vs. 0.84), while the appearance distance does not change significantly (-0.035, 95% CI $[-0.21, 0.19]$). A training-free attention bias that ties tokens to their carrier trajectory, in the spirit of flow-guided attention [6, 10], was selected on 50 clips with an earlier refiner (boundary jump -16% there) and reduces the seams by 9% on the full set (2.06 vs. 2.26; better on 364 of 526 clips), but it slightly increases flicker within chunks and leaves the readability loss in place. We therefore keep the refiner out of Banksy. The negative result is specific to our scale—1.3B parameters and at most a few hundred real video pairs, against a 14B backbone and a million training triplets for SteerVTE—but it is consistent: at this scale, a learned harmoniser trades the guarantee that motivates rendering for an appearance gain we could not measure.

Table 8. **Two appearance modules that did not pass their pre-registered criteria** (same front-end pass and evaluation job as Tab. 1). ΔApp.: paired appearance difference to B1 with 95% bootstrap CI, on the clips outside both development sets (477 clips); the criterion required the upper bound to be below 0. R1 changed 7 clips, 5 of them development clips.

	Read. ↑	Net ↑	App. ↓	ΔApp. vs. B1
Banksy (B1, final)	91.4	198	2.57	–
+ baseline layout + glyph-shape font (B1f)	91.3	197	2.61	+0.052 [+0.002, +0.106]
+ source-glyph reuse (R1)	91.4	198	2.56	+0.021 [−0.023, +0.067]
+ both (R1f)	91.3	197	2.57	+0.034 [−0.016, +0.092]

5.7. Two appearance modules that did not pass

Two of the causes of the appearance gap (Sec. 6.2) look addressable without generation: the new text ignores the orientation and the typeface of the old one, and the garbled source already contains most of the target's letters. We built one module for each and fixed their acceptance criteria before running them on the benchmark (supplement). Both were run in the same front-end pass and evaluation job as Tab. 1, so every comparison in Tab. 8 is paired.

Baseline-aligned layout and glyph-matched fonts (B1f). The first component lays the new text along the baseline of the old glyphs instead of the axis-aligned OCR quad, when two ink-based angle estimates agree within 2° and the rotation lies between 2° and 20°. The second chooses the typeface by the shapes of the source letters whose identity can be established—the multi-frame consensus reading aligned to the target—and replaces the matched font only if the best per-letter overlap is at least 0.6 (English, single row). The criteria were: readability at most one point lower, a paired appearance difference whose 95% CI lies below zero, and perceived naturalness not worse in a later human study. The text was rotated in 157 clips and the font replaced in 25; elsewhere the output equals B1. Readability is unchanged (0 vs. 1 discordant clips), but the appearance difference is +0.052 [+0.002, +0.106] on the clips outside both development sets, so the second criterion is not met and B1f is

not adopted. Under the current evaluation protocol we observe no overall benefit; its perceptual effect has not been verified. We do not read this as evidence that font or orientation alignment is ineffective, for two reasons. First, the appearance distance is glyph-agnostic by construction (colour, contrast, halo, edge softness and shading inside the strokes, Sec. 3.1): it does not see typeface similarity or text orientation, which are exactly what B1f changes. Choosing it as the criterion for these components was an error in our pre-registration, which we report rather than amend after the fact. Second, the distance is not deterministic: its ink segmentation runs k-means with random initialisation, and on the 517 clips where R1 leaves the output byte-identical to B1, the two scores agree exactly in only 28 clips, differ with a standard deviation of 0.47 per clip, and differ by $+0.021$ $[-0.020, +0.062]$ on average. B1f's difference is of the same order as this test–retest noise. Deciding whether B1f looks better requires its own pre-registered human study.

Source-glyph reuse (R1). R1 keeps the source letters that are already correct in place—with their own pixels, font, material and perspective—erases only the wrong span between them, and renders only the missing letters, in a font chosen to match the kept ones. It acts only when the evidence is strong: the number of segmented source letters equals the reading, at least half of the target letters are kept, the kept letters are read stably across frames, no letter must be inserted or deleted between kept letters, the new letters' shapes match the kept ones (overlap at least 0.65), and the new text stays on its carrier; otherwise the clip keeps the B1 output. R1 was applied to 7 of the 526 clips (6 of the 112 garbled English clips), 5 of them development clips. On the two others the appearance distance improved by -0.44 with unchanged readability, which is far too few clips to evaluate the criteria. The limiting factor is coverage: in 67 English clips the segmented letters did not match the reading, in 15 the pipeline's consensus reading of the source already equalled the target, and in 15 no font matched the kept letters well enough. With our current letter segmentation, matching and gating, reuse applies to too few clips to form a general module. This bounds the implementation, not the idea; the 20% coverage observed on a 30-clip sample before the full run did not hold on the benchmark.

Both modules are frozen and are not part of Banksy, which in Tab. 1 is B1.

A glyph-locked material generator. The appearance gap invites a learned module that keeps Banksy's glyphs and generates only their material. We built one and registered four gates for it (supplement). Training data are counterfactual pairs rendered in Blender: one scene, one material and one camera path with two strings, a garbled one and the target, so that the only difference between the two renders is



Figure 8. **The material generator on real clips** (gate R2; three hand-picked examples; Original / Banksy / generator, with the reader's transcription). Trained only on rendered pairs, the generator keeps the glyphs (their alpha is an input) but writes them with low contrast and desaturated colour on most carriers, which the appearance distance and the reader both penalise; it is closer to the original than Banksy only on metal.

the text (2,000 training pairs over 8 carrier materials, 5 text styles including extruded and metallic letters, and 41 fonts; 80 validation pairs that use the benchmark's target strings, which the training set excludes). A conditional U-Net in the carrier's UV space receives the erased background, the target glyph alpha and the other string's rendered material, and predicts the text layer's colour and alpha; the glyph alpha fixes identity, layout and topology. On held-out pairs the model closes 0.48 of the gap between a flat paste and the rendered truth (median 0.44; positive for every material, lowest on metal at $+0.33$) with a glyph IoU of 0.95, just below the registered 0.50. Applied to the 129 benchmark clips outside the development sets that both Banksy and SteerVTE repair (Fig. 8), it fails every registered criterion: its appearance distance is 3.06 against 1.99 for Banksy (paired $+1.08$, 95% CI $[+0.73, +1.45]$; closer than Banksy on 24% of the clips and, by material, only on metal), and the reader loses 20 clips (84.5% vs. 100%). The outputs keep the glyph shapes (the alpha is an input, which fixes shape but not contrast) and are low in contrast and desaturated on most carriers, which also explains the lost readings. The rendered training crops are bright (mean intensity 213 of 255) and their text differs from its surround by a median of 18 luminance levels, so a synthetic-to-real gap in photometry is the obvious suspect. A registered ablation with the frozen weights (supplement) confirms it as one cause, not the whole one: matching the predicted ink's colour statistics to the source text's on the 50 non-emissive clips moves the paired appearance difference to Banksy from +1.16 to

+0.29 [-0.32, +0.77] and recovers 8 of the 12 lost readings, but the calibrated output still reads below Banksy (88.0% vs. 100%) and stays farther from the original than a flat paste of the source's median colour (+0.79 [+0.23, +1.61]). Mapping the inputs to the training photometry instead helps the same clips and hurts the dark and emissive ones, which the training set lacks. The remainder is the model, not the domain, and we do not retrain it.

5.8. Verified selection between Banksy and SteerVTE

The two systems fail on different clips (Sec. 6.2), so a system that could tell which output is right would inherit the readability of both. We registered one before running it (supplement), with no tunable threshold. *Banksy-V* keeps the clip when the preservation gate fires; otherwise it runs SteerVTE (seed 1, then a second seed) and accepts an output whose text an independent verifier—InternVL2.5-8B with the reading protocol of Sec. 3.1—transcribes as the target in at least 4 of 5 frames; otherwise it falls back to Banksy. An accepted SteerVTE output is not used as a whole frame: only its text region is composited back into the original, with weight 1 inside the text box and up to 0.3 box heights outside it (shadow and glow), a linear fade over 0.12 box heights, and a per-channel affine colour match to the original on an annulus around the composite. The verifier never scores: readability is reported by the first reader and, on the English clips, by Pixtral, which have no part in the selection. Table 9 evaluates every row in one run (the second of two; the first, which contained only the pre-registered variants, gave the same verdicts).

Result. Banksy-V reads correctly in 96.4% of the clips, 30 vs. 6 discordant clips against Banksy (McNemar $p = 7.0 \times 10^{-5}$), with 0 originally correct clips broken and 225 net repairs (Banksy: 201; SteerVTE: 160); the third reader agrees (87.5% vs. 77.6% for Banksy and 80.3% for SteerVTE). Its appearance distance (2.67) lies between SteerVTE's (2.37) and Banksy's (3.34), because SteerVTE's text is used wherever it reads correctly—in 370 clips with the first seed and 33 with the second—and the background stays local (41.74 dB against 37.41 dB when the whole SteerVTE frame is accepted). The verifier accepted 7 outputs that the first reader judges wrong, 6 of which Banksy had right; an oracle that selects with the scoring reader itself reaches 98.1%, so the verifier closes most of the gap between the two systems. Banksy-V meets five of the six pre-registered criteria; it misses the temporal one by a small margin (flicker 1.86 vs. a limit of 1.85, jitter 8.68 vs. 8.40), which the whole-frame variant does not (1.72, 8.28): the per-frame colour match of the composite adds frame-to-frame variation. A variant with one colour match per clip, defined after this result, brings flicker within the limit (1.83) but not jitter (8.62); the remaining difference to the whole-frame variant is the re-encoding of the composited output, which the whole-frame row does not undergo: re-encoding SteerVTE's own outputs with our writer raises their flicker by 0.15 and jitter by 0.44, and the per-clip-matched composite is then indistinguishable from them (supplement).

What this does and does not show. Banksy-V is a system combination: its material comes from SteerVTE, its correctness from verification, and its locality and fallback from Banksy. It runs SteerVTE (108 s per clip on its hardware) up to twice per clip plus a 7B-parameter verifier, and it inherits SteerVTE's licence and compute. It does not make Banksy's own glyphs look more like the original; the material gap of Sec. 6.2 is unchanged for the 77 clips that still come from Banksy. What it shows is that, on this task, an independent reader is a reliable enough judge of correctness that a generative editor can be used where it is right and avoided where it is wrong.

5.9. External benchmarks

We re-ran both external benchmarks with the frozen final system (B1, the configuration of Tab. 1, run from a code snapshot) and keep the earlier runs with a superseded front end only as history (supplement). All methods of a benchmark are evaluated in one evaluation run (Tab. 10). Neither benchmark matches our task exactly, so we state what each method receives.

Editing correct text (VTE-Bench Real). The 100 real clips of SteerVTE's own benchmark [42] ask to change correct text into a different string. Banksy receives the source clip, the target string, the source string (used only to locate the text by OCR) and the benchmark's text mask as hint box; SteerVTE receives the mask, a rendering of the target and an image of the source style, following its official protocol. In 8 of the 100 clips the frozen front end does not find enough detection boxes inside the hint and ships the clip unchanged, as it would in deployment; these clips are scored as unchanged and listed in the supplement. On this task Banksy is clearly behind SteerVTE: 73.0% against 92.0% readability under the first reader (4 vs. 23 discordant clips, $p = 3.1 \times 10^{-4}$), 68.0% against 91.0% under the second ($p = 3.4 \times 10^{-5}$), and 32.0% against 47.0% under EasyOCR. It changes less outside the text (37.5 vs. 35.4 dB) and flickers and jitters less, but the unlocated clips alone cost up to 8 points, and on correct, well-formed source text the gates that make Banksy safe for repair have nothing to protect, whereas SteerVTE is trained for exactly this replacement. We report the result to delimit our claim: Banksy is a repair method for garbled generated text, not a general video text editor.

Table 9. **Verified selection between Banksy and SteerVTE** (526 clips; one evaluation run for every row, so Banksy and SteerVTE are re-evaluated here). Banksy-V keeps the clip when the preservation gate fires, otherwise takes a SteerVTE output (seed 1, then seed 2) whose text an independent verifier (InternVL2.5-8B, the protocol of Sec. 3.1) reads as the target, composited back into the original frame around the text box, and falls back to Banksy otherwise. Scored by the first reader; the verifier never scores. W/L vs. Banksy: clips correct only with the system / only with Banksy (McNemar p). Pixtral: readability on the 295 English clips under the third reader.

	Read. ↑	Fixed	Broken ↓	Net ↑	W/L vs. Banksy (p)	Appear. ↓	BG PSNR ↑	Flicker ↓	Jitter ↓	Pixtral ↑
Banksy (B1)	91.8	213	12	201	–	3.34	42.52	1.79	7.86	77.6
SteerVTE	84.0	165	5	160	29/70 (4.6×10^{-5})	2.37	36.49	1.66	8.30	80.3
Banksy-V: verified selection, local composite	96.4	225	0	225	30/6 (7.0×10^{-5})	2.67	41.74	1.86	8.68	87.5
whole-frame instead of local composite	96.0	224	1	223	30/8 (4.7×10^{-4})	2.63	37.41	1.72	8.28	87.8
seed 1 only	95.6	221	0	221	25/5 (3.2×10^{-4})	2.76	41.82	1.85	8.61	86.8
per-clip colour matching (V1.1, defined after V1)	96.2	225	1	224	30/7 (1.9×10^{-4})	2.69	41.74	1.83	8.62	87.8
Oracle selection (selector = reader; upper bound)	98.1	234	0	234	34/1 (2.1×10^{-9})	2.70	41.74	1.88	8.84	88.5

Table 10. **External benchmarks with the frozen system** (one evaluation run per benchmark). Top: VTE-Bench Real (100 clips, editing correct text); readability by the two VLM readers and EasyOCR. Bottom: VidScribe V2V (44 clips); readability on the 29 replacement and translation clips, and for the 15 deletions the share whose source text is no longer read. Clips the front end could not locate are scored unchanged (8 and 1). BG: PSNR outside the dilated text box.

VTE-Bench Real (100 clips)	Read. Qwen	Read. InternVL	Read. EasyOCR	BG PSNR	Flicker	Jitter
Original	0.0	0.0	0.0	–	1.64	5.89
SteerVTE	92.0	91.0	47.0	35.40	2.02	7.74
Banksy (frozen)	73.0	68.0	32.0	37.46	1.29	4.43

VidScribe V2V (44 clips)	Read. Qwen (29)	Read. InternVL (29)	Removed Qwen (15)	Removed InternVL (15)	BG PSNR	Flicker
Original	0.0	0.0	26.7	26.7	–	2.62
Banksy (frozen)	37.9	34.5	100.0	100.0	36.21	3.34
minimax-h3	31.0	27.6	73.3	73.3	19.75	4.98
wan-3.0	24.1	24.1	80.0	80.0	24.15	4.51
gemini-omni-1.1-flash	20.7	20.7	73.3	73.3	27.98	5.18
pixverse-v6	17.2	17.2	93.3	93.3	19.79	4.90
happyhorse	10.3	10.3	86.7	93.3	20.71	6.84
kling-3.0-omni	6.9	3.4	100.0	100.0	16.70	3.88
seedance-2.5	3.4	10.3	93.3	93.3	27.19	4.82

Commercial video models (VidScribe). VidScribe [46] releases the outputs of seven commercial video models for video-to-video text editing. We use the 44 of its 93 clips whose operation is a replacement, translation or deletion and whose source text can be located by OCR (27 style-only and 9 insertion clips are excluded by operation, 13 by localisation); Banksy receives the source clip, the target string, the source string and the located box, whereas the commercial models were prompted with an instruction and receive no box. The frozen front end does not locate 1 clip, which is shipped unchanged. Deletions have an empty target, so readability is undefined for them: we report readability on the 29 replacement and translation clips and, for the 15 deletions, the share in which the source string is no longer read (similarity below 0.5 in at least 4 of 5 frames). On the 29 clips, Banksy reads correctly in 37.9% under the first reader (34.5% under the second), ahead of the strongest commercial model (minimax-h3, 31.0%) without significance (6 vs. 4 discordant clips, $p = 0.75$) and significantly ahead of 3 of the seven models ($p < 0.05$). It removes the source text in 100.0% of the deletions. The commercial models regenerate the whole clip and lower the PSNR outside the text region to 16.7–28.0 dB, against 36.2 dB for Banksy, and they flicker more; their text is, however, closer in appearance to the original (lowest appearance distance: wan-3.0). Absolute readability is low for every method because most VidScribe targets are long phrases.

6. Analysis and limitations

6.1. Where Banksy fails

Breaking correct text. Banksy breaks 12 of the 283 originally correct clips (SteerVTE: 6). In all of them, the evaluation reader transcribes the original correctly while the pipeline's OCR reads it correctly in at most 2% of the frames (neon italics, extruded or condensed letters), so the preservation gate cannot fire. Inspecting the outputs (Fig. 9, last three rows) gives four failure types: new text larger than its sign, so that letters fall off the carrier (4 clips); old text left visible next to the new text, because a second row



Figure 9. **Failure modes of Banksy** (hand-picked to cover each type; SteerVTE shown for reference). The first three keep the target readable but degrade appearance; the last three come from clips whose original was already correct and that Banksy broke.

was not detected or the erasure was incomplete (5); wrong colour or contrast on lightboxes and extruded letters (2); and stacked two-row text on an LED screen that is still misread (1). In one of these clips the reader's verdict on the

original is itself doubtful: the original shows a garbled four-character string, which the reader transcribes as 电影院 in every frame.

Readable but not faithful. The more common failure does not affect readability (Fig. 9, first three rows): erasure leaves a visible patch, or the new glyphs are flat where the old ones were outlined, extruded or glowing. These cases are frequent in the random sample of Fig. 4 and explain the appearance distance in Tab. 1.

6.2. The appearance gap to generative editing

Among the 145 garbled clips that both methods repair, SteerVTE's text is closer to the original appearance in 70% (median appearance distance 0.93 vs. 1.79), so the gap is not an artefact of SteerVTE leaving garbled text untouched. We see four causes, in decreasing order of visibility.

- **Rendering instead of generation.** Banksy can reproduce colour, shading and, with SAT, glow; it cannot reproduce extrusion, bevels, tube geometry or texture inside the strokes, which a generator synthesises naturally.
- **A small font library.** The library holds 33 Latin and 8 Chinese families, and the generator's typefaces are invented. In practice, 26 Latin families are selected, but for Chinese only 3, and Noto Sans SC alone renders 67% of the edited Chinese clips; the result reads correctly but looks uniform. Choosing the font by the shapes of the identifiable source letters (Sec. 5.7) changed the font in only 25 clips.
- **Axis-aligned layout.** The letter band inherits the orientation of the OCR quad, which is axis-aligned for most detectors, so on tilted carriers the new text can look skewed relative to the sign (Fig. 1a); the perspective estimate of an earlier front end was too fragile to keep (supplement), and laying the text along the source baseline (Sec. 5.7) could not be shown to help with our appearance metric, which does not see orientation.
- **Erasure traces.** LaMa fills the hole once per clip; on textured or shaded carriers the fill is visible as a patch.

Two observations suggest that the gap is closable without giving up correctness. First, the garbled source already contains most of the target's characters: in garbled English clips, a median of 88% of the target letters occur in the reader's transcription of the original (Chinese: 67%). Reusing these glyphs—with their own font, colour, lighting and perspective—and synthesising only the missing ones would extend the "change only what is wrong" principle from words to characters; our first implementation (Sec. 5.7) shows that the bottleneck is reliable letter segmentation and matching, which currently admits too few clips. Second, the two methods fail on different clips: the share of clips that at least one of them repairs is 97.3%. A combination that accepts a generative edit only when independent OCR verifies it, and otherwise falls back to deterministic repair, is a natural next step; its verifier must be strict, since accepting a wrong edit costs correctness, whereas rejecting a right one costs only appearance.

6.3. Other limitations

The benchmark is generated by one family of models and its prompts ask for short sign texts; a second generator (CogVideoX-5B, 360 clips from the same prompts) is part of our plan but has not been evaluated yet, and long texts such as menus or user interfaces are out of scope. Readability is measured by a VLM reader; it favours large, high-contrast text, which may flatter rendered glyphs, and it can transcribe a garbled original as the intended word (Sec. 6.1). A pre-registered second reader from another model family confirms the advantage over SteerVTE, with a smaller margin that is significant for Chinese but not for English alone, and a third reader limited to English finds the methods level once its invented additions are discounted (Sec. 5.1); readability claims therefore depend on the reader, and we report all of them. The appearance distance is glyph-agnostic and has a test–retest noise of 0.47 per clip (Sec. 5.7): it supports large differences such as the gap to SteerVTE and the effect of SAT, but not small ones. The human study on SAT used four raters after quality control; a registered blind comparison of Banksy with SteerVTE on material naturalness, artifacts and temporal stability was not conducted for lack of raters, so the appearance comparison with SteerVTE rests on the appearance distance and on visual examples. Finally, the preservation gate is safe but has low recall, because it relies on a conventional OCR that rarely reads stylised originals. Outside repair, on editing correct text (VTE-Bench), the frozen system is clearly behind SteerVTE (73.0% vs. 92.0%, Sec. 5.9), partly because its front end fails to locate 8 of the 100 source texts.

7. Conclusion

Repairing garbled text in generated videos is not the same problem as editing text: the source is a near miss, it is often already right, and everything around it should survive. Banksy treats it as a sequence of evidence-gated decisions—leave the clip alone, lay the text out on the old glyphs or on the tracked box, transport the old appearance or not—around a rendering step whose spelling cannot drift. This makes it more reliable than the strongest generative editor on our benchmark and confines its changes to the text, but its glyphs still look rendered rather than generated. Our analysis and two pre-registered attempts indicate what closing that gap without giving up correctness requires: letter segmentation and matching reliable enough to reuse the characters that the generator already drew correctly, an appearance measure that sees glyph shape and orientation, and generative edits that are accepted only when they can be verified—a step we take in Sec. 5.8, where verified selec-

tion between the two systems repairs more clips than either and breaks none, at the cost of running the generative editor.

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Banksy: Evidence-Gated Repair of Garbled Text in AI-Generated Videos

Supplementary Material

A. Benchmark details

Generation. Clips were generated with the text-to-video mode of VACE-14B from 120 prompts, each asking for a scene with a sign of a given material that reads a quoted target string, with ten seeds per prompt; 1,178 of the 1,200 generations were usable (832×464, 49 frames, 15 fps). An OCR screen confirmed the requested text in 24% of them (English 21%, Chinese 27%). The 526 benchmark clips were selected as garbled by an EasyOCR-based screen (selection rule to be documented); our evaluation reader judges 53.8% of them correct (Tab. 1), so the benchmark contains repair and do-nothing cases.

Targets and composition. The 40 target strings are: `BUS STOP` (25 clips), `FRESH BREAD` (23), `BOOK STORE` (22), `PHARMACY` (22), `COFFEE SHOP` (21), `TEA HOUSE` (20), `NO PARKING` (19), `FLOWER MARKET` (19), `SALE 50%` (18), `NIGHT MARKET` (18), 面包房 (17), 小心地滑 (17), 邮局 (16), `ICE CREAM` (15), 停车场 (15), 电影院 (15), 书店 (14), `RECORDS` (13), 出口 (13), 洗衣店 (13), 咖啡店 (12), 茶馆 (12), 欢迎光临 (11), 营业中 (11), `OPEN` (10), `EXIT` (10), 理发店 (10), `GALLERY` (9), 花店 (9), 公园 (9), 火锅 (9), `BAKERY` (8), 夜市 (8), `HOTEL` (7), 便利店 (7), 药房 (7), `LAUNDRY` (6), 水果店 (6), `CINEMA` (5) and `WELCOME` (5). Table 11 gives the material × language composition and the readability of each method per group.

Table 11. Composition of the benchmark and readability (%) per group (same run as Tab. 1).

Group	EN	ZH	Original	SteerVTE	Banksy
neon	47	46	58.1	89.2	97.8
signboard	74	12	64.0	77.9	93.0
wood	20	34	63.0	90.7	92.6
metal	22	34	71.4	94.6	96.4
glass	52	24	34.2	77.6	84.2
paper	55	41	47.9	80.2	87.5
screen	25	40	43.1	83.1	89.2
English (295)			62.0	84.1	91.5
Chinese (231)			43.3	84.0	91.3

Development data. We disclose every clip that was inspected during development. (i) Twelve clips ("dev12") were used to tune the instance front end and the SAT oracle; 24 random clips ("held24") were not used for tuning, but screenshots of four of them were viewed during diagnosis and held24 informed the SAT margin, so they are not strictly held out. (ii) The 53 SAT-enabled clips were used for the SAT human study and then for the failure diagnosis of Sec. 5.4; results of the padded canvas on them are development results. (iii) Forty clips were inspected while developing the selective-editing components; these are the clips excluded in the lower block of Tab. 1 and the right block of Tab. 6: `g001_en_neon_s1 g006_en_paper_s3 g007_en_glass_s6 g008_en_signboard_s0 g010_en_neon_s9 g011_en_wood_s1 g020_en_glass_s5 g020_en_glass_s7 g022_en_neon_s1 g024_en_metal_s0 g025_en_paper_s4 g028_en_signboard_s3 g029_en_screen_s2 g036_en_neon_s6 g045_en_signboard_s8 g046_en_neon_s2 g046_en_neon_s4 g046_en_neon_s5 g049_en_wood_s5 g052_en_screen_s4 g057_en_neon_s6 g057_en_neon_s7 g059_en_metal_s2 g067_zh_wood_s9 g071_zh_neon_s0 g076_zh_neon_s5 g076_zh_neon_s8 g083_zh_neon_s1 g084_zh_neon_s0 g084_zh_neon_s3 g093_zh_neon_s7 g098_zh_neon_s1 g107_zh_neon_s2 g107_zh_neon_s9 g110_zh_neon_s0 g110_zh_neon_s4 g110_zh_neon_s5 g110_zh_neon_s7 g110_zh_neon_s8 g116_zh_screen_s8.` The main conclusions do not change when they are removed. (iv) Nine clips were inspected while developing source-glyph reuse (Sec. 5.7): three development clips (`g026_en_signboard_s7 g055_en_signboard_s6 g024_en_metal_s2`) and six clips of the originally registered 30-clip evaluation sample that were viewed while debugging and therefore moved to the development set (`g008_en_signboard_s8 g025_en_paper_s7 g026_en_signboard_s0 g029_en_screen_s6 g035_en_glass_s8 g054_en_glass_s0`). Table 8 excludes both (iii) and (iv).

B. Evaluation details

Reader. Qwen2.5-VL-7B transcribes the hint-box crop (15% padding) of frames 0, 12, 24, 36 and 48. A frame is exact if its normalised edit similarity to the target is at least 0.999 after normalisation (whitespace removed, lowercased, traditional Chinese folded to simplified); a clip is correct if at least 4 of 5 frames are exact. An earlier draft of the pre-registration described this as "at least 60% of frames"; this was a documentation error, the code was fixed before the run, and the decisions are unaffected.

Evaluation hygiene. In an earlier phase, the reader produced garbled transcriptions (a repeated token) for about 8.5% of the inputs on one cluster (x86, an older version of the inference library) and none on another (aarch64, newer version). Two refiners had been read on the clean system and their baselines on the faulty one, which inflated a comparison by 16.6 points; after re-reading everything in one

job the advantage disappeared (Appendix I). Since then, every table is produced by re-reading all of its rows in a single job, and we check transcriptions for garbage. Re-reading identical outputs in a second job changes readability by about one point, which is why we treat differences of that size as ties: when the 99 unchanged originals of the two-row re-run were read again, 75 received identical transcriptions and 98 the same verdict. The front end itself is not bit-reproducible across runs (GPU OCR boxes, LaMa on different devices); comparing two passes of identical code gives no significant difference in appearance or flicker but a 0.46 dB difference in background PSNR, so we only compare variants from the same pass.

Appearance distance. Features per frame (CIELAB, inside the box): ink colour difference to the original, ink–ground contrast difference, halo excess normalised by contrast, boundary gradient normalised by contrast (edge softness), and standard deviation of lightness inside the eroded strokes (inner shading). The original compared with itself scores about 0.27, the estimation floor. Each feature is divided by its median over all method–clip pairs of the run, so values are comparable only within one evaluation job. The ink segmentation inside the estimator uses k-means with random initialisation (OpenCV's k-means++ without a fixed seed), so scores are not bit-reproducible: in the run of Tab. 1, 517 clips were scored twice on byte-identical outputs (B1 and R1 where reuse did not apply), and the two scores agreed exactly in 28 clips, with a per-clip standard deviation of the difference of 0.47 and a mean difference of $+0.021$ $[-0.020, +0.062]$ (the pair B1f/R1f: -0.020). The evaluation code is frozen, so we report this noise instead of changing the estimator; differences of a few hundredths in a run mean are within it.

C. Implementation details

Reference frame. Among the frames whose OCR score is at least 90% of the maximum, the one closest to the middle of the clip.

Gates. Preservation gate: complete target read in $\geq$85% of the frames, wrong readings in $\leq$5%, reading inside the hint box, extra boxes in $\leq$10% of the frames, and the concatenated reading of all boxes inside the hint box equal to the target in $\geq$90% of the correct frames. Localization gate: $\geq$3 letters ($\geq$2 for Chinese), CIELAB distance between ink inside the letter band and inter-letter ground $\geq$40, letter height between 0.3 and 1.0 of the OCR box height, no fallback row grouping; otherwise the box policy is used (168 of the 480 clips that Banksy edits in the final run, most often because OCR grouped the text into rows that the instance estimate could not confirm; in an earlier pass, the reasons were the row grouping for 134, letter height for 67, letter count for 40 and contrast for 36 clips, with overlaps). Appearance gate: emissive text (ink brighter than ground, or glow detected on the carrier) and a cross-validated error ratio below 0.75 against the flat fill in both directions.

Letter height and polarity. Latin letter height is the median height of letter-like components (solidity $\geq$0.85); Chinese letter height is taken from the ink distribution. Ink and ground are separated by k-means with $k \in \{2, 3\}$ scored by letter morphology (component heights and holes), with a fixed seed.

Font library. 33 Latin and 8 Chinese families from Google Fonts, variable fonts expanded over their weights. In the final run, 26 Latin families and 3 Chinese families were selected (Noto Sans SC for 67% of the edited Chinese clips, Noto Serif SC for 20%, ZCOOL XiaoWei for 13%).

SAT. The operator of Eq. (1) is fitted per distance bin (signed distance in letter heights, support up to 0.6 outside the strokes) by least squares on the reference frame and applied to every frame through the carrier homography. The flat fill is the ink colour of the box policy. Cross-validation fits on the left half of the text row and tests on the right half, and vice versa.

Runtime. On an NVIDIA A40, the earlier front end took about 20 s per clip and the learned refiner about 40 s per clip; SteerVTE reports 108 s per 49-frame clip on its hardware. The frozen final pipeline processed the 100 VTE-Bench clips in about 8.6 s per clip and GPU on GH200 nodes (two parallel workers, 7 min 11 s after a model-loading time of 28 s); this is a wall-clock estimate from the run logs, a controlled per-clip measurement on the benchmark is pending.

D. Pre-registration of the selective-editing study

The study was registered before the run in an internal document with dated revisions. **Criteria.** B1: fewer broken clips, false skips $\leq$2% of the skipped clips, paired readability not lower than B0 by more than 0.5 points. B2: collateral change decreases ($p < 0.05$), readability not lower than B0 by more than one point, and the human artifact question favours B2. B3: infeasible flags confirmed by manual inspection; results reported with infeasible clips counted as failures (main) and excluded (reference). Components are attributed separately before B4 is considered. **Revisions during development** (on the development clips only, before the run) concerned the B2 glow model and the B3 boundary cues. **Revisions during the run** (before any result was seen, recorded at the time): B3 feasibility was

changed to use the character density of the source, measured on the best reading over all frames; B1 received its fifth condition after two false skips were found while checking decisions (a correct word above garbled small text; a near-miss reading split across boxes). All 526 clips were then re-run with the final code. **Round two** (defined after the results, before any human data): a colour-completion variant of B2 and a whole-word-subset variant of B3 were tested on nine and six diagnosis clips, respectively, and were not run at scale. **Excluding infeasible clips** (reference): readability B1 92.0, B3 91.8; after manual review, replacing the non-confirmed B3 flags by B0 outputs gives 90.3%.

E. Pre-registration of the appearance modules

Both modules of Sec. 5.7 were registered before the full run in the same internal document format. **R1 criteria** (on the clips where R1 applies, outside its development clips): at most one clip less readable than B1; paired appearance difference with 95% CI upper bound below zero; in a blind study, R1 preferred in more than half of the non-tied judgements with no more artifacts; coverage reported. Only all three together count as success. **B1f criteria** (all clips and the clips outside the development sets): paired readability difference not below -1 point; paired appearance difference with CI upper bound below zero; naturalness not worse in a later human study; meeting the appearance criterion alone is not sufficient. **Revisions during development**, on the three R1 development clips only: multi-frame consensus reading with per-letter stability, word-initial and word-final insertion and deletion, letter-by-letter layout at the source spacing instead of horizontal compression, a font-consistency gate on the kept letters, SAT fitted on the kept letters, and baseline-aligned segmentation. **Contamination.** The first run on the registered 30-clip evaluation sample applied R1 to 6 clips; these were then viewed while debugging, which revealed that LaMa continued the kept letters into the erased span (fixed by taking the middle background from the full-erase background), that appended text could leave its carrier (fixed with the carrier-extent estimate of B3), and that compressed letters merged into blobs (fixed by rendering letters individually). Because they had been inspected, the six clips were moved to the development set, and the evaluation of R1 was replaced by the full 526-clip pass with the final code; a first full pass started with the earlier code was cancelled and its outputs were discarded. **Outcome.** B1f fails the appearance criterion; R1 cannot be evaluated because it applies to two clips outside its development set. The choice of a glyph-agnostic appearance distance as the criterion for font and orientation changes was a design error; we report the registered criterion unchanged.

F. Pre-registration of the material generator

Registered before the pilot. **Gates.** R0 (data): at least 90% of 40 pilot pairs render, and in 10 inspected pairs the two renders differ only in the text. R1 (synthetic counterfactual transfer, on held-out pairs whose strings do not occur in training): the model closes at least 0.5 of the gap between a flat paste and the rendered truth (the 1.3B refiner of Sec. 5.7: -0.06); glyph IoU ≥ 0.85; reported per material, with metal, neon and glass not negative. R2 (real clips, zero-shot): the benchmark clips that both systems repair, outside the development sets, with B1's erased background, B1's glyph alpha and the original as the only inputs; one evaluation run together with Banksy and SteerVTE; appearance distance against B1 with CI upper bound below zero and mean at most -0.30, readability not more than one point below B1, BG PSNR ≥ 41 dB. R3 (all 526 clips): the R2 criteria plus the temporal limits and the second reader. No benchmark clip is used for training or model selection. **Deviation.** R1 closed 0.482 against the registered 0.50 (IoU 0.954; every material positive); the gate rule called for a diagnosis before R2, and R2 was run anyway as a diagnostic, recorded as such before its result was seen. **Outcome.** R2 fails all three criteria (Sec. 5.7); R3 was not run. **Photometric ablation** (registered after the R2 result, before its run). The frozen weights are applied to the same 129 clips with (P1) one per-clip, per-channel affine map that takes the reference crop to the training-set mean and spread, applied to both inputs and inverted exactly on the predicted text layer; (P2) the predicted ink's per-channel mean and spread matched to the source text's, on non-emissive clips only (the front end's glow flag, or source ink brighter than its surround by more than 30 luminance levels), with gains clipped to $[0.5, 2]$; and both (P12). The apply step now composites the erased crop only inside the erase mask (the R2 run replaced the whole window, which is why its BG PSNR is 38.8 dB for the model and the flat paste alike); the flat paste and the unaligned model (P0) are regenerated with the same code, and every row is evaluated in one run. Decision rules, in order: (a) a variant meets the R2 criteria against B1: photometry is the main cause; (b) its appearance distance is below P0's (CI upper bound below zero) with readability not worse: photometry is one cause and the remainder lies in the model; (c) no variant improves on P0: photometry is not the cause and the synthetic-to-real wording is withdrawn. Independently, (d) P2 against the flat paste asks whether the model's spatial structure adds anything beyond colour statistics. Retraining with photometric augmentation is deferred unless (a) or (b) holds. **Result** (Tab. 12; one run, 129 clips, of which 50 are non-emissive under the registered rule and 79 are skipped by P2). P2 and P12 fall under (b): on the non-emissive clips the appearance difference to P0 is -0.87 [-1.73, -0.27], with 8 readings gained and 2 lost ($p = 0.109$), and the difference to Banksy

Table 12. Photometric domain-alignment ablation of the material generator (frozen v0 weights, the 129 R2 clips, one evaluation run; appearance distance normalised within this run and not comparable to other tables). P1 maps a clip's reference crop to the training statistics before the model and undoes the map on its output; P2 matches the predicted ink's colour statistics to the source text's on non-emissive clips; the flat paste is the geometry pipeline alone. The original is the garbled input.

Row	Read ↑	Appear. ↓	BG ↑	Flicker ↓	Jitter ↓
All 129 R2 clips					
Original	0.0	–	–	2.07	9.00
SteerVTE	100.0	1.21	36.1	1.87	8.66
Banksy (B1)	100.0	1.65	42.2	1.75	7.32
Flat paste (geometry only)	89.1	1.58	41.4	2.70	9.94
Generator v0 (P0, as registered)	86.8	2.42	41.3	2.63	9.45
+ input alignment (P1)	87.6	2.52	41.3	2.51	9.09
+ output calibration (P2)	91.5	2.09	41.3	2.65	9.57
+ both (P12)	88.4	2.45	41.3	2.53	9.18
Non-emissive clips where P2 acts (50)					
Original	0.0	–	–	1.50	6.92
SteerVTE	100.0	1.67	36.1	1.26	6.52
Banksy (B1)	100.0	1.82	42.3	1.25	5.92
Flat paste (geometry only)	88.0	1.32	41.5	2.53	9.41
Generator v0 (P0, as registered)	76.0	2.97	41.5	2.41	8.56
+ input alignment (P1)	82.0	2.11	41.5	2.45	8.89
+ output calibration (P2)	88.0	2.11	41.5	2.47	8.89
+ both (P12)	84.0	2.00	41.5	2.49	9.04

shrinks from +1.16 [+0.67, +1.69] to +0.29 [-0.32, +0.77]; (a) is not met, because readability stays below Banksy's (88.0% vs. 100%). P1 falls under (c) on all clips (+0.09 [-0.31, +0.42]): it helps the non-emissive clips (-0.87 [-1.72, -0.28]) and hurts the dark and emissive ones, as expected of a training set whose crops have a mean intensity of 213 of 255 and whose text differs from its surround by a median of 18 luminance levels (71% of the training text by less than 30). Rule (d) gives a stronger answer than the registered one: P2 is not level with the flat paste but worse (+0.79 [+0.23, +1.61]), and the flat paste, which keeps only the source's median colour, is as close to the original as Banksy on this glyph-agnostic distance (1.58 vs. 1.65) while reading worse (89.1%) and flickering more (its per-frame warp is not Banksy's compositing). We keep the negative result, attribute part of it to photometry, and do not retrain: the registered gate measures colour statistics, which a constant colour satisfies, so a better generator would first need a criterion that rewards structure and a human study, neither of which this paper has. With the erase-mask compositing, BG PSNR is 41.3 dB for every generated row.

G. Second reader and external re-runs

Both were registered before their full runs. **Second reader.** The outputs of Tab. 1 (original, SteerVTE, Banksy) are re-read with the identical crop, frames, prompt, decoding (greedy, at most 40 new tokens) and scoring by a reader from a different model family; each crop is first resized with the first reader's own pixel budget, then passed to the second reader's standard preprocessing. Pixtral-12B (Mistral) was the first candidate; in a 12-clip smoke test it produced unrelated text for every Chinese crop of every method, including outputs the first reader read correctly, so it is used for the 295 English clips only. InternVL2.5-8B-MPO (InternViT and internLM2.5, independent of both) reads all 526 clips. Both are run in the software environment of the first reader so that the decoded crops are identical, and both are reported regardless of the outcome; no threshold or component of Banksy may be changed on the basis of these reads. The registered interpretation: the readability advantage is called reader-robust only if, under InternVL, Banksy is more readable than SteerVTE with McNemar $p < 0.05$ on all 526 clips. **External re-runs.** The frozen Banksy (code snapshot of the run of Tab. 1) is run on VTE-Bench Real and VidScribe V2V with the inputs stated in Sec. 5.9; clips that the front end cannot locate are shipped unchanged, written through the same reader and writer as all other outputs (so their background PSNR reflects re-encoding rather than a bit-identical copy), and listed. The earlier runs with a superseded front end are kept only as history (Appendix I). **Execution.** The cluster of the first reader was unavailable for days, so the second readers, the single-run re-evaluation of Tab. 6 and the external benchmarks ran on a second cluster with V100 GPUs. Its decoded and resized evaluation crops are byte-identical to the first environment's (450 of 450 crops checked); only the GPU and the CUDA build differ, and the first reader re-reading the outputs of Tab. 1 there reproduces 1576 of 1578 verdicts. Reads were split over clips across identical GPUs and merged; V100s have no fused bf16 attention, so InternVL and Pixtral were run with batch size 1. The temporal metric received a guard against division by zero on static clips, which first occurred on VTE-Bench; values are unchanged wherever the earlier code returned one. Decisions and collateral change for Tab. 6 were computed once on the first cluster from the same merged outputs.

H. Pre-registration of the verified selection

Registered before the run. **Candidates**: the frozen Banksy (B1), SteerVTE seed 1 and seed 2 (the seed-2 outputs existed already). **Verifier**: InternVL2.5-8B with the evaluation protocol (same crop, 5 frames, at least 4 exact); it was chosen over the pipeline's own OCR because EasyOCR accepts only 193 of the 442 SteerVTE outputs the first reader

judges correct, which caps a hybrid at 92.8%. **Decision order**: preservation gate → original; verifier reads seed 1 → its local composite; else seed 2 → its composite; else B1. No threshold is tuned. **Local compositing**: weight 1 inside the per-frame text box and up to 0.3 box heights outside, linear fade over 0.12 box heights, per-channel affine colour match on the annulus 0.42–0.92 box heights outside; the three parameters were set by eye on three development clips before the run. **Scoring**: the first reader (Qwen2.5-VL-7B) and, on English, Pixtral-12B; the verifier never scores. **Criteria**: readability at least 2 points above B1 with McNemar $p < 0.05$; no more broken clips than B1; BG PSNR ≥ 41 dB; appearance distance at least 0.30 below B1; flicker ≤ 1.85 and jitter ≤ 8.40 (no worse than the worse parent); Pixtral readability not more than 1 point below B1's. **Outcome**: five criteria met; the temporal criterion missed (flicker 1.86, jitter 8.68). **Post-hoc variant** (defined after seeing this): the colour match is estimated per frame but applied as one per-clip median gain and offset, so that it cannot vary over time; it was composited and evaluated in a second, separate run together with every other row, which is the run of Tab. 9; in that run the variant passes the flicker limit (1.83) and misses jitter (8.62 vs. 8.40), so the criterion remains unmet for both variants. The first run (preregistered variants only) gave 96.4% readability and the same five-of-six verdict. **Diagnosis.** The composites are written by the same video writer as every Banksy output, whereas SteerVTE's files are its own H.264 encodes. Re-encoding SteerVTE's outputs with our writer, with no other change, raises their flicker by $+0.150$ [$+0.141$, $+0.159$] and their jitter by $+0.436$ [$+0.402$, $+0.468$] (paired over the 526 clips); the per-clip-matched composite is then indistinguishable from the re-encoded SteerVTE output ($+0.000$ [-0.002, $+0.002$] flicker, $+0.004$ [-0.005, $+0.014$] jitter), and the per-frame-matched composite adds $+0.035$ and $+0.083$. The temporal criterion was set against SteerVTE's own encode, so it compares two encoders rather than two systems; we leave the verdict as registered and report the cause.

I. Additional results

Diagnosing the instance front end. On the 379 clips where an earlier gated variant accepted an instance, forcing the box policy's colour pipeline (88.4%) or a letter height of at least 0.6 box heights (88.1%) did not close the gap to the box policy (92.3%); replacing the box erase did not help either (87.6%), whereas replacing the tracking state machine by carrier tracking did (90.8 vs. 91.3%). A refinement with row splitting from the ink profile, a perspective trapezoid fitted to letter tops and bottoms (only with $\geq$6 homogeneous letters and consistent fits) and shadow-aware erasure fixed several stacked and perspective cases visually but lowered readability on aligned clips by 1.6 points and was not kept.

Material-conditioned rendering. We also tried rendering models per material (glow for neon and screens, bevel and shadow for metal, paint following the texture for wood, paper and signboards, translucency for glass), selected by the prompt label or by a VLM prediction. A wrong material model hurts consistently (appearance $+0.28$, $p < 0.001$; worse on six of seven materials), but the right one helps clearly only on screens (-0.49, $p < 0.001$) and hurts on glass ($+0.30$). The prompt label agrees with the visual material in only 60.5% of the clips (the generator often renders signboards and screens as glowing text). An automatic pairwise realism judge (the same VLM, both orders averaged) scored 0.49–0.51 for every pair and could not discriminate. These models were dropped; SAT replaces them for emissive carriers.

Occluders. Carrier anchoring assumes that nothing passes in front of the text. On 90 garbled clips with synthetic occluders of known alpha (30 each crossing, fully covering for 13 frames, or covering half of the text), the temporally anchored pipeline erased or over-painted occluders more than naive pasting (occluder damage 10.3 vs. 5.4 grey levels). An occlusion map from a readable-frame median background with per-pixel adaptive thresholds reduced the damage to 3.4 (-66%) but also lowered readability on visible frames (67% vs. 73%) because of false positives on strokes. Occlusion handling is therefore not part of Banksy, and clips with occluders remain a limitation.

Learned refiner. Table 13 summarises the refiner study on the 296-clip garbled subset (window-level readability, all rows read in one job) and on 1,500 held-out image pairs. "Closed" is $1 - \sum|\hat{I} - I| / \sum|P - I|$ over the text region, where P is the paste and I the original (0: equal to the paste; 1: perfect).

Table 13. Learned refiner: data scale and supervision. Video: 296 garbled clips (Lab dist.: mean CIELAB distance to the original in the original ink region; lower is closer); image: 1,500 held-out pairs (our-font paste; "closed†" with the original's own glyph pasted).

	Video		Held-out image pairs		
	Read.	Lab dist.	Read.	Closed	Closed†
Paste	80.7	53.9	81.3	0	0
Refiner, 0 image pairs	78.4	51.1	68.9	−0.121	–
10k pairs	75.3	51.7	51.9	−0.059	–
100k pairs	76.4	51.8	55.5	−0.064	−0.491
1M pairs	76.4	53.1	52.7	−0.058	–
Glyph-consistent, 100k	77.7	52.8	77.0	−0.026	+0.049
Glyph-consistent, 1M	78.7	52.7	76.5	−0.029	–
+ video pairs, street ×4	79.7	52.0	76.7	−0.029	–

Combining with a generative editor. An earlier version of our system used SteerVTE as an editor inside a verify-and-select loop: skip clips whose text is already correct, try SteerVTE first, accept its output only if EasyOCR reads the target, otherwise try our earlier refined rendering, and finally a second SteerVTE seed. On the 526 clips it reached 91.1% readability, 82.2% on the garbled clips and kept 98.6% of the correct clips correct, against 84.0%, 67.8% and 97.9% for a single SteerVTE run (earlier front end; the reader-selected best of two SteerVTE seeds reaches 89.4%). This supports the complementarity noted in Sec. 6.2; its appearance was not evaluated.

Historical external runs (superseded front end). Before the final system was frozen, both external benchmarks were run with the earlier box-policy front end. On VTE-Bench Real, SteerVTE reached 92% readability and our earlier pipeline 82% (8 vs. 18 discordant clips, $p = 0.076$; EasyOCR: 49% vs. 44%). On the 44 VidScribe clips, the earlier pipeline reached 29.5% (naive paste) and 27.3% (box policy), ahead of the strongest commercial model (minimax-h3, 18.2%) without significance ($p = 0.29$) and significantly ahead of five of the seven models ($p \leq 0.031$); the commercial models lowered the PSNR outside the text region to 19.8–27.9 dB against 37.6–37.9 dB for our outputs. These numbers were scored over all 44 clips, including the 15 deletions whose empty target cannot be read, and are superseded by Sec. 5.9.

External front-end failures. The frozen front end did not find enough detection boxes inside the hint box, and shipped the clip unchanged, for 8 VTE-Bench clips—clip 18 (`EMILY` → `FRONT`), clip 20 (`OKI` → `Bed`), clip 30 (`The` → `PAY`), clip 38 (`FACE REC` → `NEW YEAR`), clip 53 (`Ly COLD CENTER` → `Happy New Year`), clip 64 (`U.S.N` → `HER`), clip 77 (`Today's` → `POLICE`), clip 85 (`HD` → `BY`)—and for 1 VidScribe clip, video_064 (replace, `NICE TO MEET YOU` → `GOOD MORNING`). These clips are scored as the unchanged source, re-encoded by the same writer as every other output so that their background PSNR is not a bit-identical copy; the earlier box-policy front end filled such clips with the static hint box instead.

Neon structure reconstruction. On the 23 SAT-enabled neon clips we explored neon-specific operators—keeping the sign frame out of erasure and SAT, rebuilding double-line tube geometry, and restricting the rendered text to the word visible in the source. A non-blind preview judged three of six examples good and three faulty (old content showing through, a changed font style, and invented text); after fixes, this line was paused in favour of selective editing.

J. Human studies

SAT study. All 53 SAT-enabled clips; each trial shows the paste-only and the SAT version left and right of the original (full frame plus an enlarged text row), with the side randomised per clip (SAT on the left in 26 of 53). Raters, who know the target, answer separately: which version's material is closer to the original, which has fewer artifacts, and which flickers more (ties allowed). Five raters completed the study; the exclusion rule and its sensitivity analysis are described in Sec. 5.4. Analyses use win rates with ties excluded, two-way bootstrap over raters and clips, clip-level sign tests, Fleiss' κ and Gwet's AC1.

Banksy vs. SteerVTE (not conducted). A blind comparison on 60 clips (42 that both methods repair, stratified by material, and 18 from the other outcome cells; questions on material naturalness, artifacts and temporal stability, with a quality-control rule fixed in advance) was registered and its page prepared, but it was cancelled before any data were collected for lack of raters. The earlier selective-editing study page (B1 vs. SteerVTE and B2 vs. B0) likewise collected no data.

K. Failure cases

The 12 originally correct clips broken by Banksy in the run of Tab. 1, by type: text larger than its carrier (`BOOK STORE` on neon, `HOTEL`, `GALLERY`, 面包房); old text left visible next to the new text (`BUS STOP`, `FLOWER MARKET` ×2, `TEA HOUSE`, 电影院); wrong colour or contrast (`WELCOME` on a lightbox, 书店 on extruded letters); and stacked two-row text still misread on an LED screen (`SALE 50%`). Eleven of them were also broken in the selective-editing pass; three clips broken there are not broken here, and one is new, which illustrates the run-to-run variation of a pipeline whose OCR is not bit-reproducible.